

# **Cours de Réseaux et Services**

## **UE :2103**

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# Plan

- I. Couche physique et média**
- II. Couche de liaison de données**
- III. Couche réseau**
- IV. Couche de transport**
- V. Couche application**
- VI. Sécurité**

# Plan

## **V. Couche application**

**1. Système de noms de domaines**

**2. Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP**

**3. Gestion de réseau: SNMP**

**4. Multimédia**

# Couche Application

## *Application Layer*

### Objectives

The application layer enables the user, whether human or software, to access the network. It provides user interfaces and support for services such as electronic mail, file access and transfer, access to system resources, surfing the world wide web, and network management.

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**The application layer is responsible for providing services to the user.**

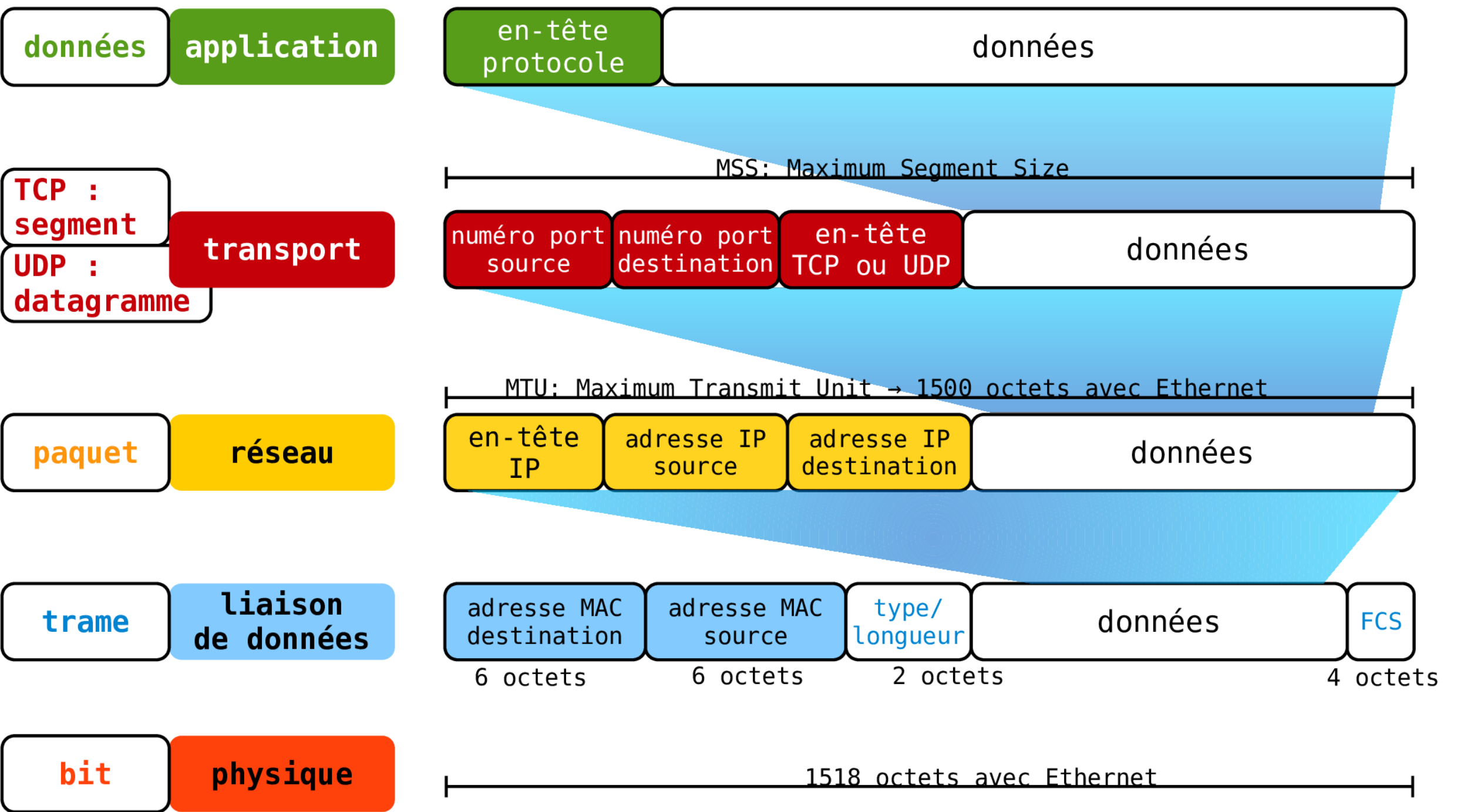
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In this part, we briefly discuss some applications that are designed as a client/server pair in the Internet. The client sends a request for a service to the server; the server responds to the client.

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**Part 6 of the book is devoted to the application layer  
and the services provided by this layer.**

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# Couche Application

## Network Models



| TCP/IP Model         | OSI Model          | Protocols         |
|----------------------|--------------------|-------------------|
| Application layer    | Application layer  | FTP, HTTP, Telnet |
|                      | Presentation layer | JPEG, MPEG        |
|                      | Session layer      | NFS, SQL, PAP     |
| Transport layer      | Transport layer    | TCP, UDP          |
| Internet layer       | Network layer      | IPv4, IPv6        |
| Network Access Layer | Data Link Layer    | ARP, CDP, STP     |
|                      | Physical layer     | Ethernet, Wi-Fi   |

<https://www.ad-net.com.tw/osi-model-tcp-ip-network-models-must-concept-understand-move-deeper-networking-adventures/>

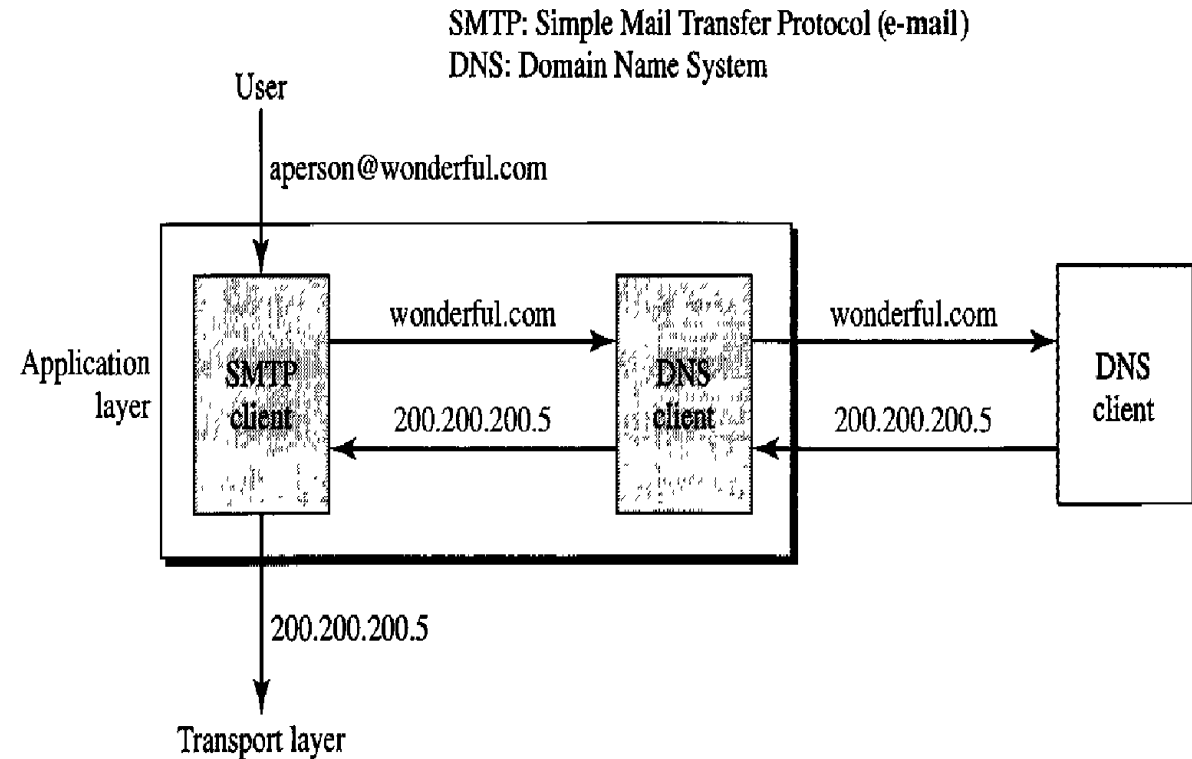
# Systeme de noms de domaines (DNS)

There are several applications in the application layer of the Internet model that follow the client/server paradigm. The client/server programs can be divided into two categories: those that can be directly used by the user, such as e-mail, and those that support other application programs. The Domain Name System (DNS) is a supporting program that is used by other programs such as e-mail.

Figure 25.1 shows an example of how a DNS client/server program can support an e-mail program to find the IP address of an e-mail recipient. A user of an e-mail program may know the e-mail address of the recipient; however, the IP protocol needs the IP address. The DNS client program sends a request to a DNS server to map the e-mail address to the corresponding IP address.

# Système de noms de domaines (DNS)

Figure 25.1 *Example of using the DNS service*





# Systeme de noms de domaines (DNS)

## NAME SPACE

To be unambiguous, the names assigned to machines must be carefully selected from a name space with complete control over the binding between the names and IP addresses. In other words, the names must be unique because the addresses are unique. A **name space** that maps each address to a unique name can be organized in two ways: flat or hierarchical.

### Flat Name Space

In a **flat name space**, a name is assigned to an address. A name in this space is a sequence of characters without structure. The names may or may not have a common section; if they do, it has no meaning. The main disadvantage of a flat name space is that it cannot be used in a large system such as the Internet because it must be centrally controlled to avoid ambiguity and duplication.

# Systeme de noms de domaines (DNS)

## Hierarchical Name Space

In a **hierarchical name space**, each name is made of several parts. The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on. In this case, the authority to assign and control the name spaces can be decentralized. A central authority can assign the part of the name that defines the nature of the organization and the name of the organization. The responsibility of the rest of the name can be given to the organization itself. The organization can add suffixes (or prefixes) to the name to define its host or resources. The management of the organization need not worry that the prefix chosen for a host is taken by another organization because, even if part of an address is the

# Systeme de noms de domaines (DNS)

same, the whole address is different. For example, assume two colleges and a company call one of their computers *challenger*. The first college is given a name by the central authority such as *fhda.edu*, the second college is given the name *berkeley.edu*, and the company is given the name *smart.com*. When these organizations add the name *challenger* to the name they have already been given, the end result is three distinguishable names: *challenger.fhda.edu*, *challenger.berkeley.edu*, and *challenger.smart.com*. The names are unique without the need for assignment by a central authority. The central authority controls only part of the name, not the whole.

# Systeme de noms de domaines (DNS)

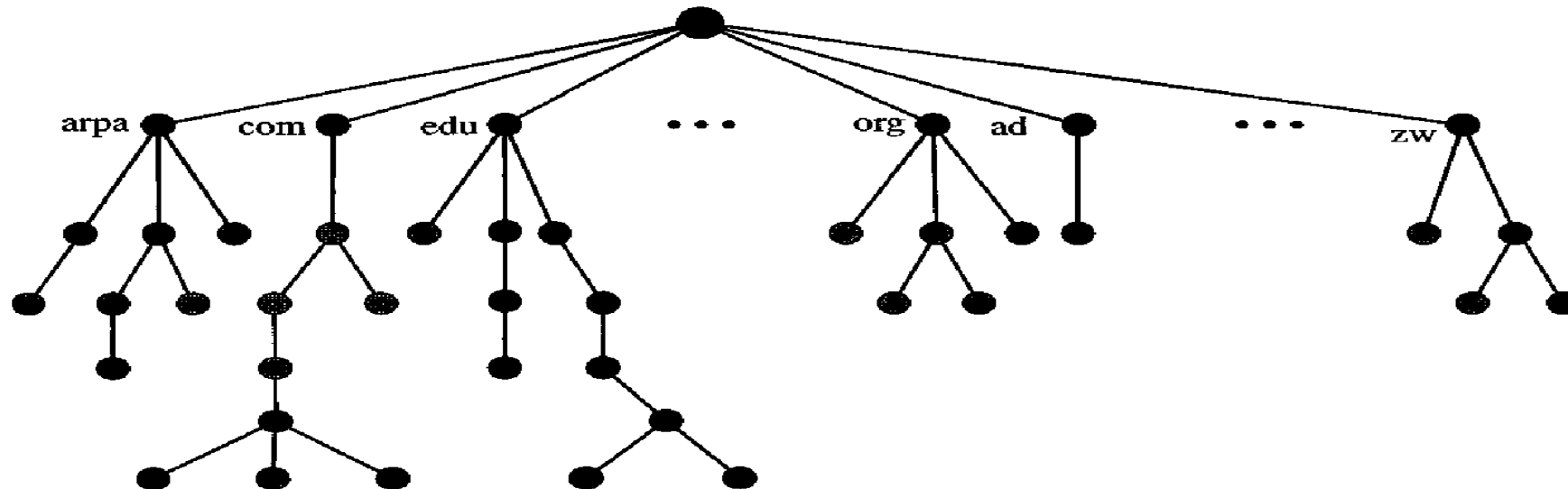
## DOMAIN NAME SPACE

To have a hierarchical name space, a **domain name space** was designed. In this design the names are defined in an inverted-tree structure with the root at the top. The tree can have only 128 levels: level 0 (root) to level 127 (see Figure 25.2).

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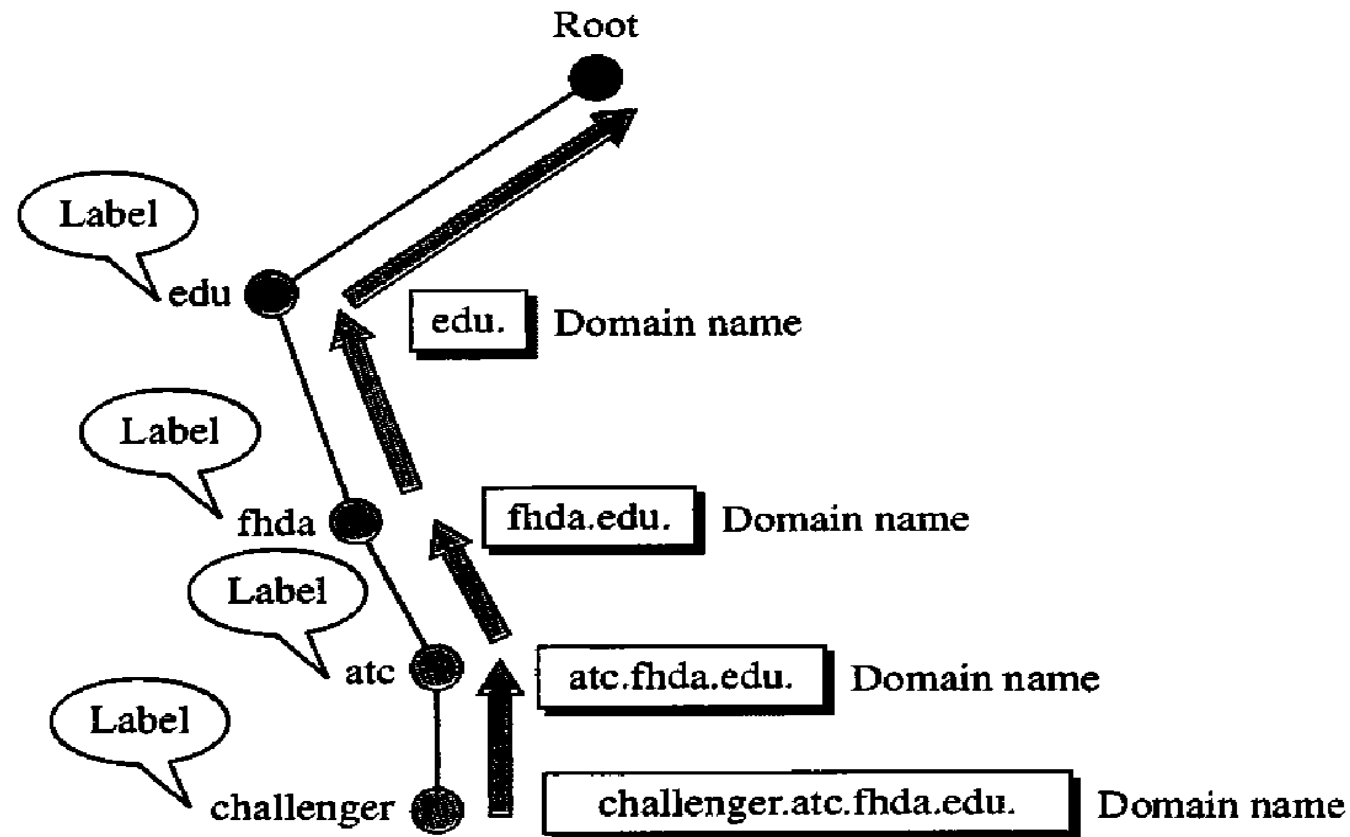
**Figure 25.2** *Domain name space*

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# Systeme de noms de domaines (DNS)

Figure 25.3 *Domain names and labels*



# Système de noms de domaines (DNS)

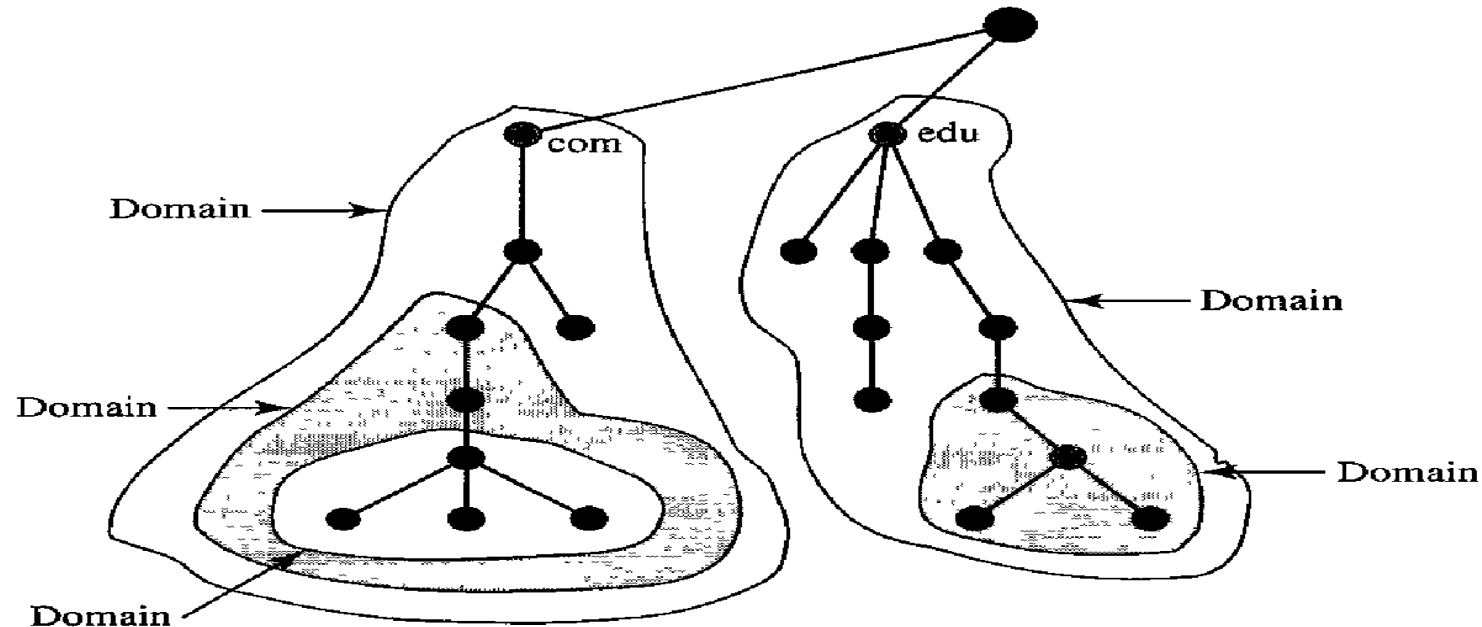
## Domain

A **domain** is a subtree of the domain name space. The name of the domain is the domain name of the node at the top of the subtree. Figure 25.5 shows some domains. Note that a domain may itself be divided into domains (or **subdomains** as they are sometimes called).

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**Figure 25.5** *Domains*

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# Systeme de noms de domaines (DNS)

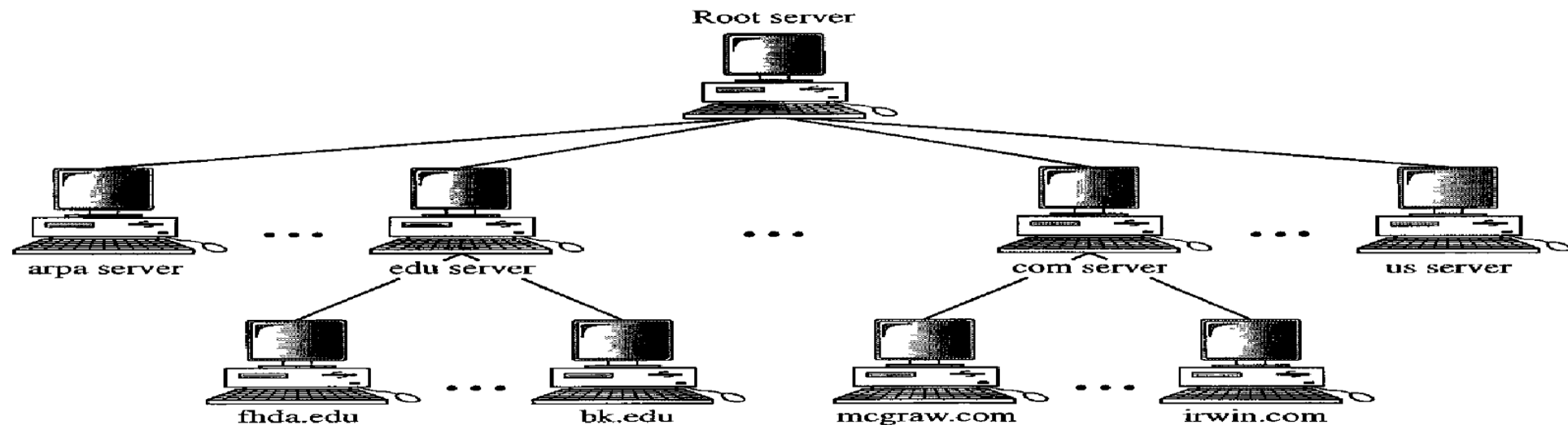
## Hierarchy of Name Servers

The solution to these problems is to distribute the information among many computers called **DNS servers**. One way to do this is to divide the whole space into many domains based on the first level. In other words, we let the root stand alone and create as many domains (subtrees) as there are first-level nodes. Because a domain created in this way could be very large, DNS allows domains to be divided further into smaller domains (subdomains). Each server can be responsible (authoritative) for either a large or a small domain. In other words, we have a hierarchy of servers in the same way that we have a hierarchy of names (see Figure 25.6).

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**Figure 25.6** *Hierarchy of name servers*

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# Systeme de noms de domaines (DNS)

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## Root Server

A **root server** is a server whose zone consists of the whole tree. A root server usually does not store any information about domains but delegates its authority to other servers, keeping references to those servers. There are several root servers, each covering the whole domain name space. The servers are distributed all around the world.



# Systeme de noms de domaines (DNS)

## Primary and Secondary Servers

DNS defines two types of servers: primary and secondary. A **primary server** is a server that stores a file about the zone for which it is an authority. It is responsible for creating, maintaining, and updating the zone file. It stores the zone file on a local disk.

A **secondary server** is a server that transfers the complete information about a zone from another server (primary or secondary) and stores the file on its local disk. The secondary server neither creates nor updates the zone files. If updating is required, it must be done by the primary server, which sends the updated version to the secondary.

The primary and secondary servers are both authoritative for the zones they serve. The idea is not to put the secondary server at a lower level of authority but to create redundancy for the data so that if one server fails, the other can continue serving clients. Note also that a server can be a primary server for a specific zone and a secondary server for another zone. Therefore, when we refer to a server as a primary or secondary server, we should be careful to which zone we refer.

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**A primary server loads all information from the disk file; the secondary server loads all information from the primary server. When the secondary downloads information from the primary, it is called zone transfer.**

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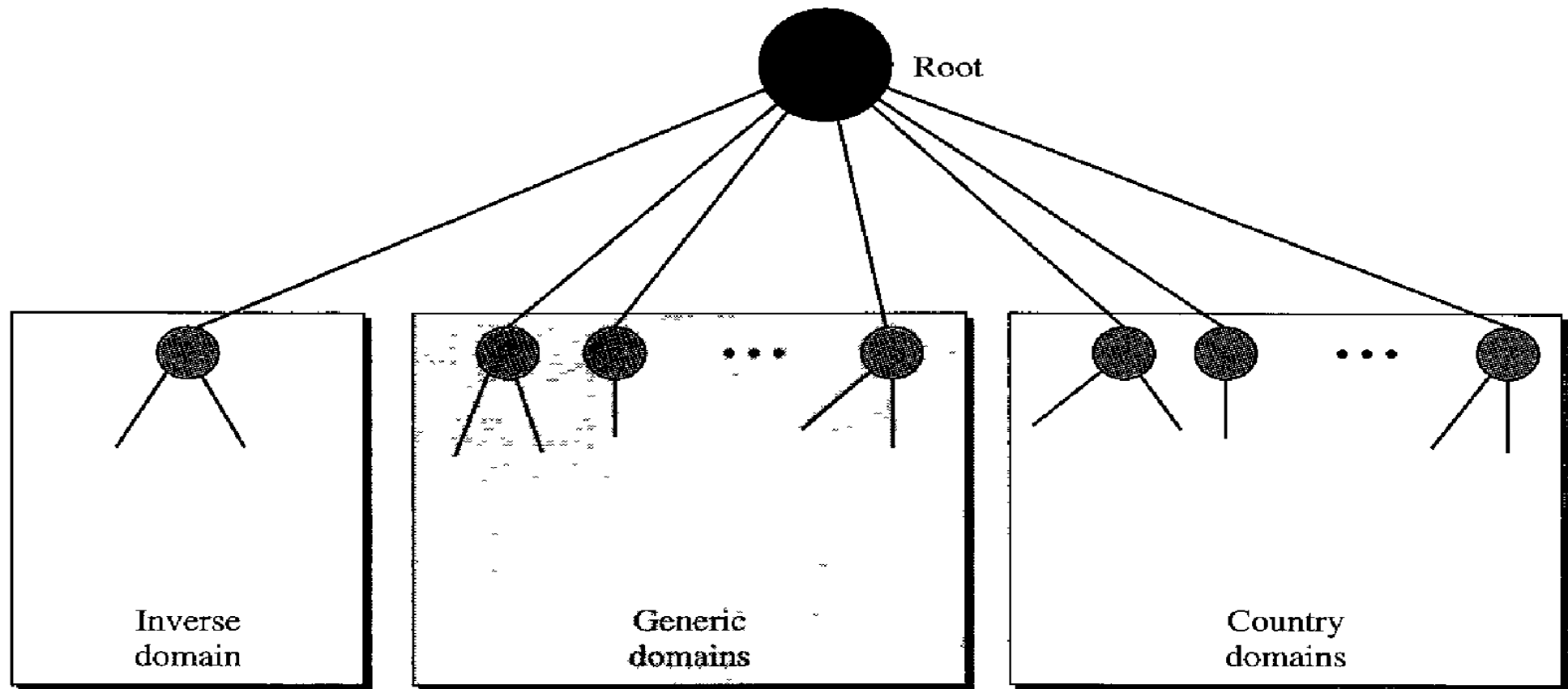
# Systeme de noms de domaines (DNS)

## DNS IN THE INTERNET

DNS is a protocol that can be used in different platforms. In the Internet, the domain name space (tree) is divided into three different sections: generic domains, country domains, and the inverse domain (see Figure 25.8).

# Systeme de noms de domaines (DNS)

**Figure 25.8** *DNS used in the Internet*



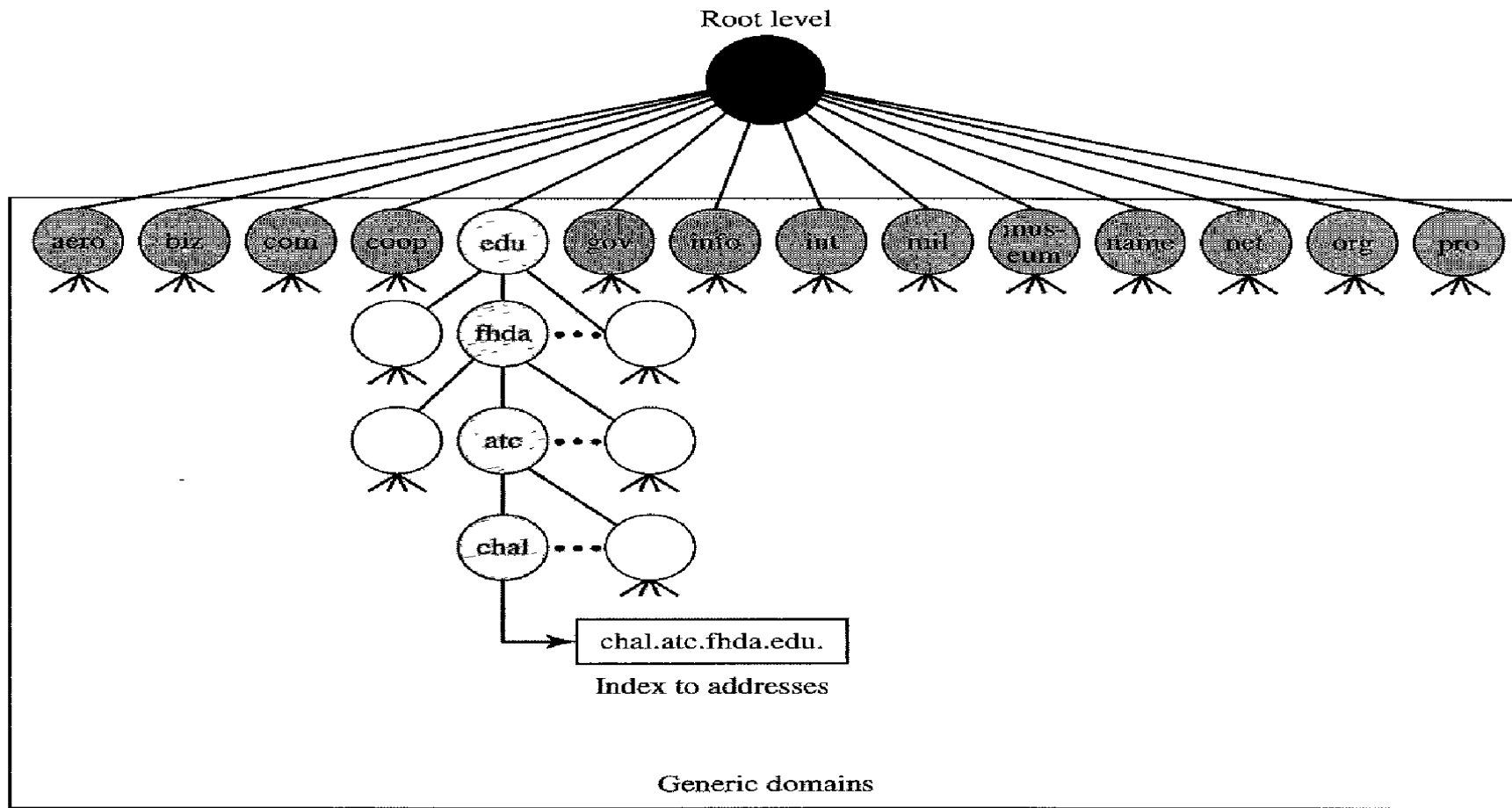
# Systeme de noms de domaines (DNS)

## Generic Domains

The **generic domains** define registered hosts according to their generic behavior. Each node in the tree defines a domain, which is an index to the domain name space database (see Figure 25.9).

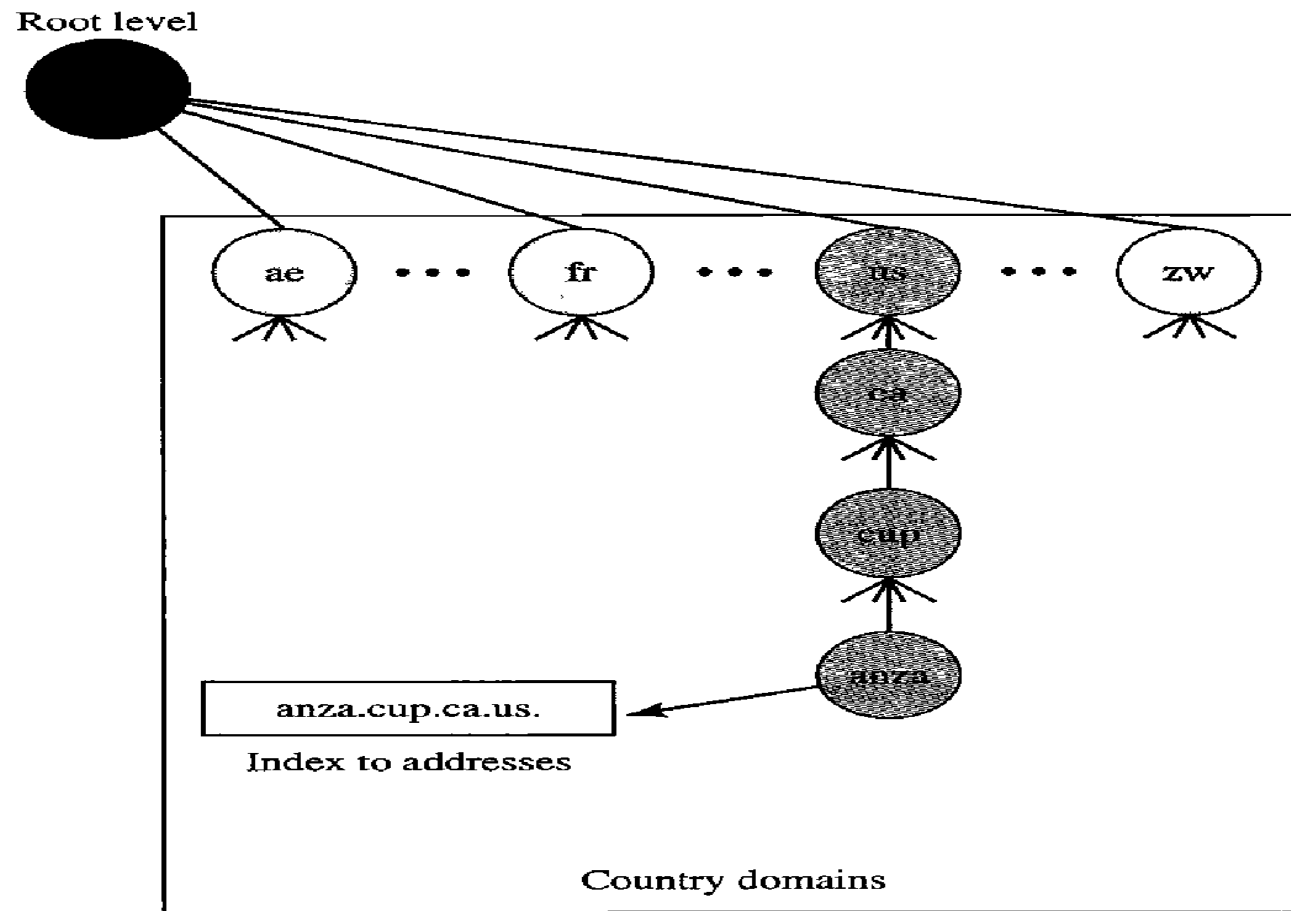
# Systeme de noms de domaines (DNS)

Figure 25.9 Generic domains



# Systeme de noms de domaines (DNS)

**Figure 25.10** *Country domains*



# Systeme de noms de domaines (DNS)

## RESOLUTION

Mapping a name to an address or an address to a name is called *name-address resolution*.

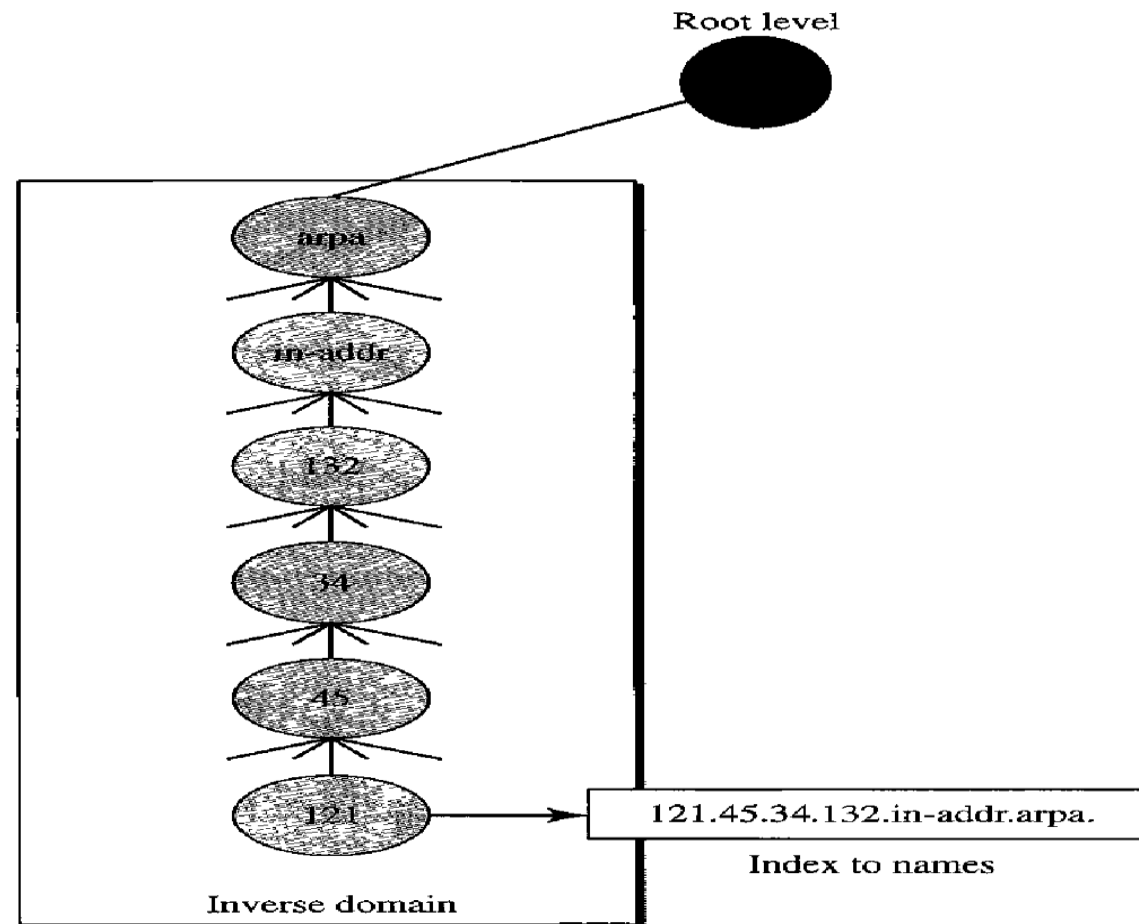
### Resolver

DNS is designed as a client/server application. A host that needs to map an address to a name or a name to an address calls a DNS client called a **resolver**. The resolver accesses the closest DNS server with a mapping request. If the server has the information, it satisfies the resolver; otherwise, it either refers the resolver to other servers or asks other servers to provide the information.

After the resolver receives the mapping, it interprets the response to see if it is a real resolution or an error, and finally delivers the result to the process that requested it.

# Systeme de noms de domaines (DNS)

**Figure 25.11** *Inverse domain*





# Systeme de noms de domaines (DNS)

## Mapping Names to Addresses

Most of the time, the resolver gives a domain name to the server and asks for the corresponding address. In this case, the server checks the generic domains or the country domains to find the mapping.

If the domain name is from the generic domains section, the resolver receives a domain name such as “*chal.atc.fhda.edu*.” The query is sent by the resolver to the local DNS server for resolution. If the local server cannot resolve the query, it either refers the resolver to other servers or asks other servers directly.

If the domain name is from the country domains section, the resolver receives a domain name such as “*ch.fhda.cu.ca.us*.” The procedure is the same.

# Systeme de noms de domaines (DNS)

## Mapping Addresses to Names

A client can send an IP address to a server to be mapped to a domain name. As mentioned before, this is called a PTR query. To answer queries of this kind, DNS uses the inverse domain. However, in the request, the IP address is reversed and the two labels *in-addr* and *arpa* are appended to create a domain acceptable by the inverse domain section. For example, if the resolver receives the IP address 132.34.45.121, the resolver first inverts the address and then adds the two labels before sending. The domain name sent is “121.45.34.132.in-addr.arpa.” which is received by the local DNS and resolved.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## REMOTE LOGGING

In the Internet, users may want to run application programs at a remote site and create results that can be transferred to their local site. For example, students may want to connect to their university computer lab from their home to access application programs for doing homework assignments or projects. One way to satisfy that demand and others is to create a client/server application program for each desired service. Programs such as file transfer programs (FTPs), e-mail (SMTP), and so on are currently available. However, it would be impossible to write a specific client/server program for each demand.

The better solution is a general-purpose client/server program that lets a user access any application program on a remote computer; in other words, allow the user to log on to a remote computer. After logging on, a user can use the services available on the remote computer and transfer the results back to the local computer.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## TELNET

In this section, we discuss such a client/server application program: TELNET. TELNET is an abbreviation for *TErminaL NETwork*. It is the standard TCP/IP protocol for virtual terminal service as proposed by the International Organization for Standards (ISO). TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.

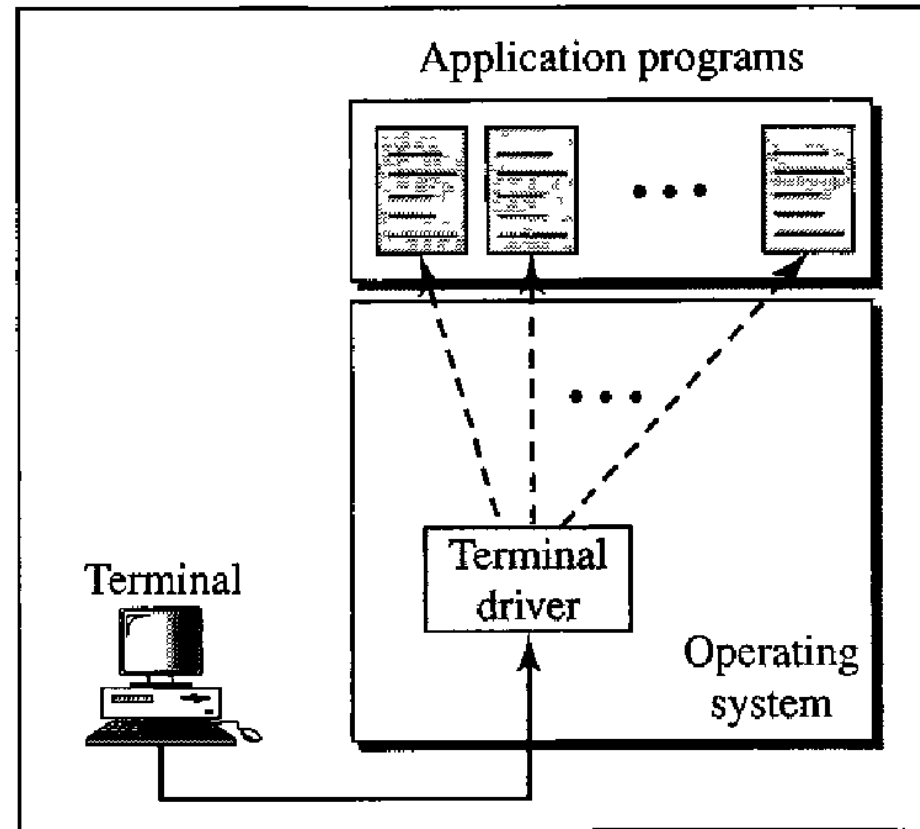
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**TELNET is a general-purpose client/server application program.**

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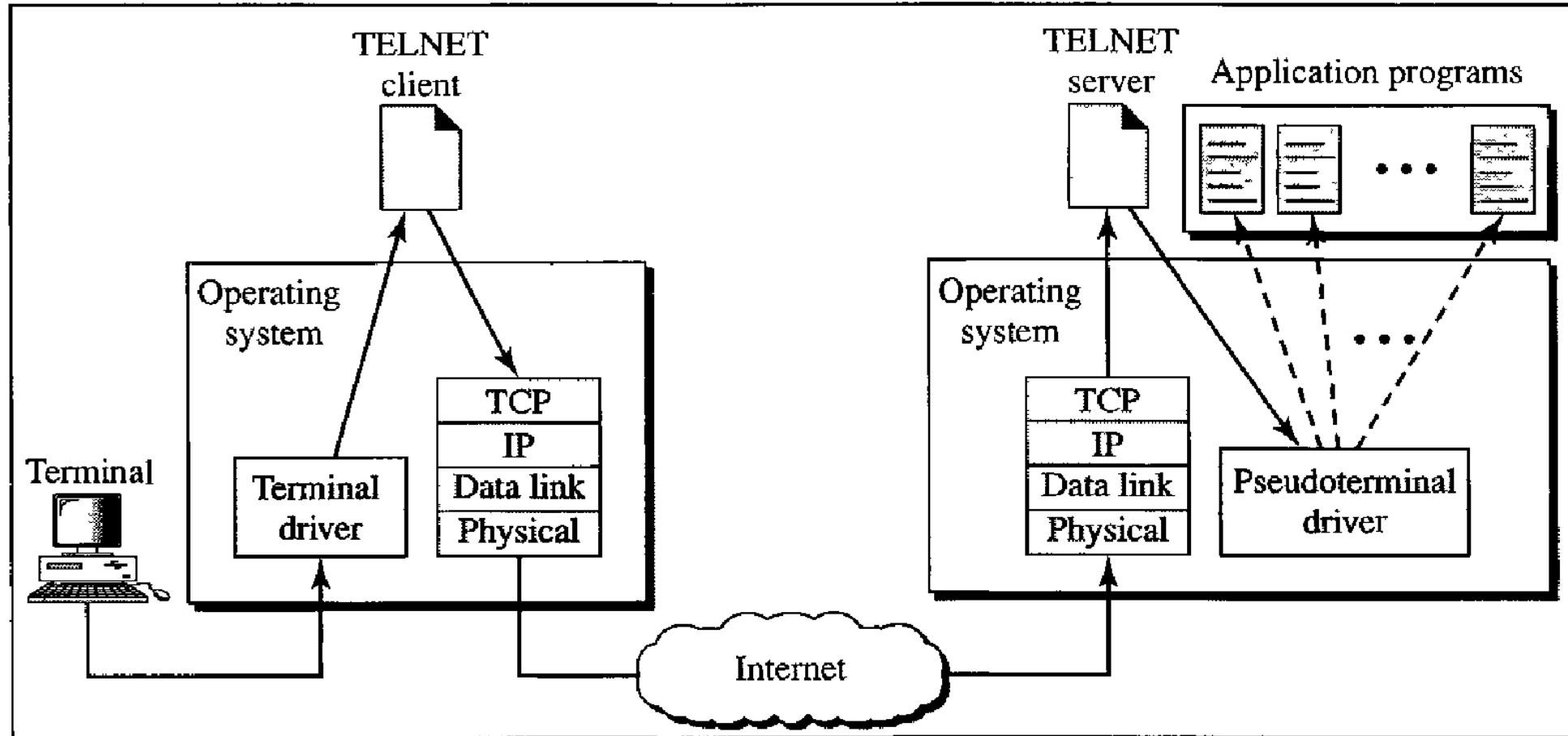
# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 26.1 *Local and remote log-in*



a. Local log-in

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP



b. Remote log-in

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

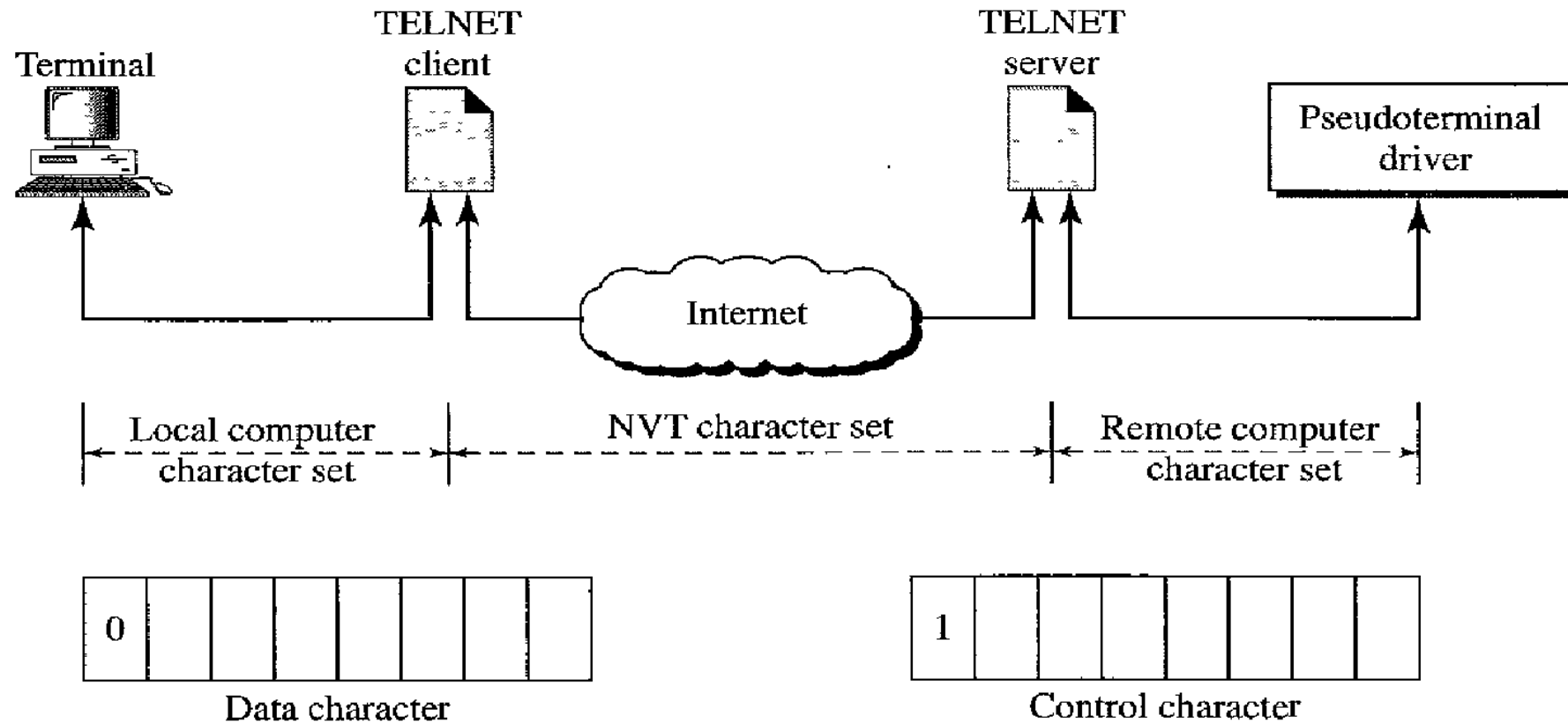
## *Network Virtual Terminal*

The mechanism to access a remote computer is complex. This is so because every computer and its operating system accept a special combination of characters as tokens. For example, the end-of-file token in a computer running the DOS operating system is Ctrl+z, while the UNIX operating system recognizes Ctrl+d.

We are dealing with heterogeneous systems. If we want to access any remote computer in the world, we must first know what type of computer we will be connected to, and we must also install the specific terminal emulator used by that computer. TELNET solves this problem by defining a universal interface called the **network virtual terminal (NVT)** character set. Via this interface, the client TELNET translates characters (data or commands) that come from the local terminal into NVT form and delivers them to the network. The server TELNET, on the other hand, translates data and commands from NVT form into the form acceptable by the remote computer. For an illustration of this concept, see Figure 26.2.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 26.2 *Concept of NVT*





# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## ELECTRONIC MAIL

One of the most popular Internet services is electronic mail (e-mail). The designers of the Internet probably never imagined the popularity of this application program. Its architecture consists of several components that we discuss in this chapter.

At the beginning of the Internet era, the messages sent by electronic mail were short and consisted of text only; they let people exchange quick memos. Today, electronic mail is much more complex. It allows a message to include text, audio, and video. It also allows one message to be sent to one or more recipients.

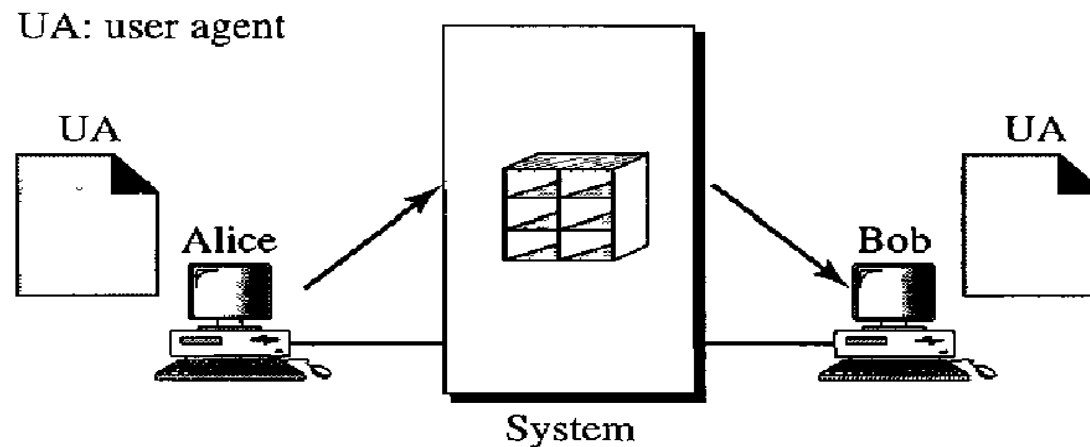
In this chapter, we first study the general architecture of an e-mail system including the three main components: user agent, message transfer agent, and message access agent. We then describe the protocols that implement these components.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## Architecture

To explain the architecture of e-mail, we give four scenarios. We begin with the simplest situation and add complexity as we proceed. The fourth scenario is the most common in the exchange of email.

**Figure 26.6** *First scenario in electronic mail*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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**When the sender and the receiver of an e-mail are on the same system,  
we need only two user agents.**

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## User Agent

The first component of an electronic mail system is the user agent (UA). It provides service to the user to make the process of sending and receiving a message easier.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

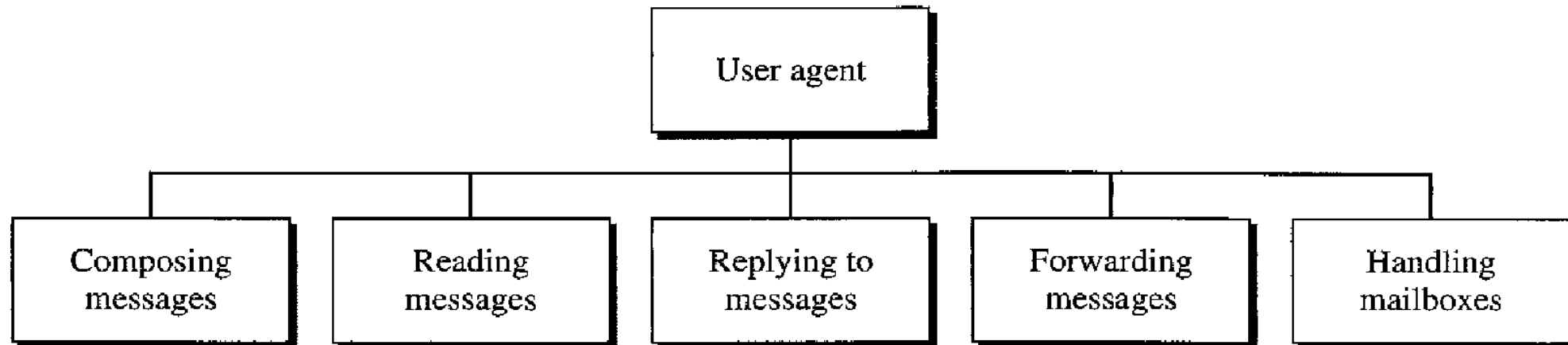
## *Services Provided by a User Agent*

A user agent is a software package (program) that composes, reads, replies to, and forwards messages. It also handles mailboxes. Figure 26.11 shows the services of a typical user agent.

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**Figure 26.11** *Services of user agent*

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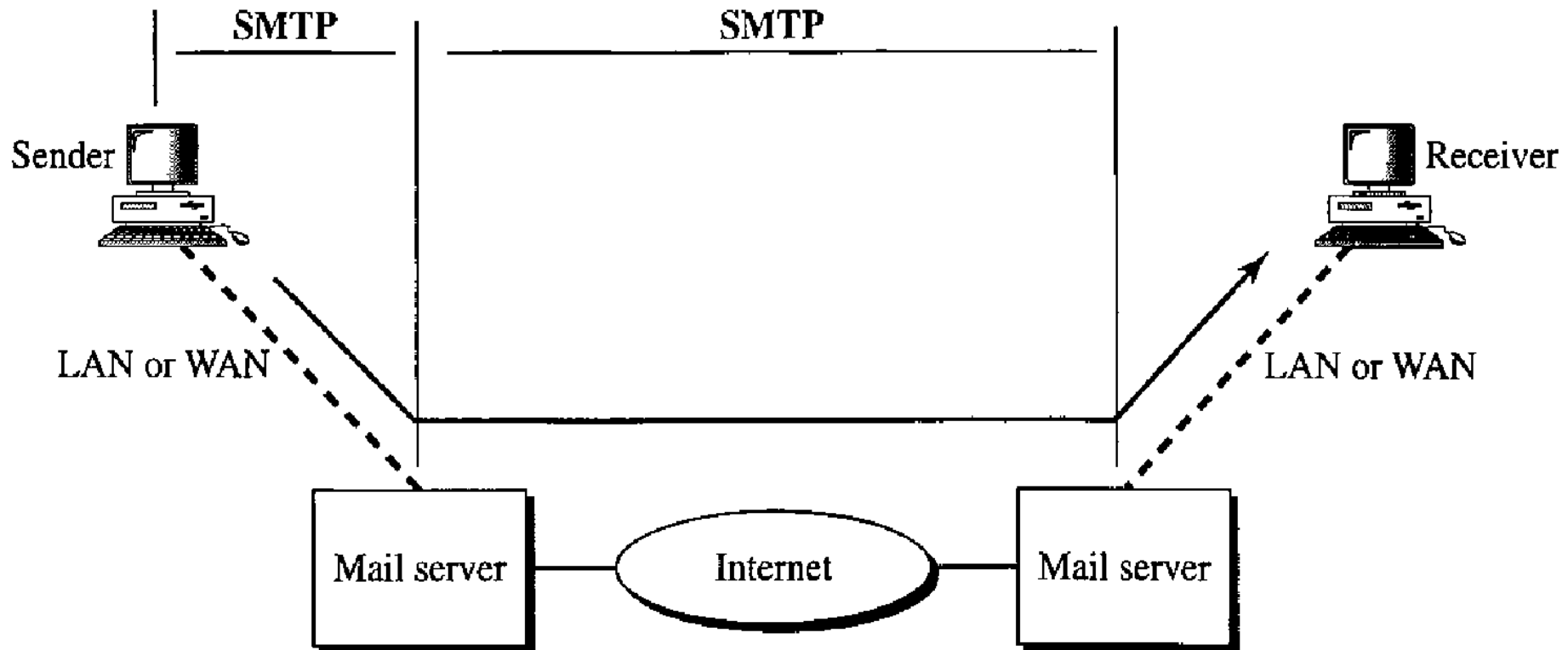
# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## Message Transfer Agent: SMTP

The actual mail transfer is done through message transfer agents. To send mail, a system must have the client MTA, and to receive mail, a system must have a server MTA. The formal protocol that defines the MTA client and server in the Internet is called the **Simple Mail Transfer Protocol (SMTP)**. As we said before, two pairs of MTA client/server programs are used in the most common situation (fourth scenario). Figure 26.16 shows the range of the SMTP protocol in this scenario.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 26.16 *SMTP range*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## Message Access Agent: POP and IMAP

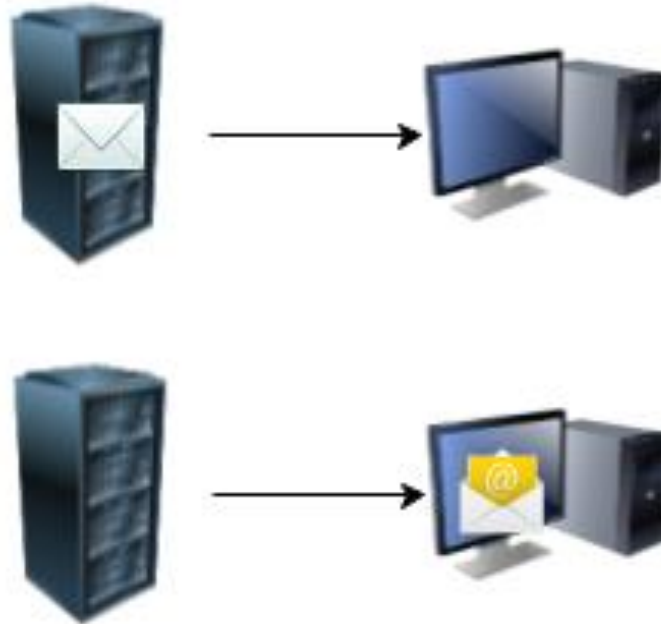
The first and the second stages of mail delivery use SMTP. However, SMTP is not involved in the third stage because SMTP is a *push* protocol; it pushes the message from the client to the server. In other words, the direction of the bulk data (messages) is from the client to the server. On the other hand, the third stage needs a *pull* protocol; the client must pull messages from the server. The direction of the bulk data is from the server to the client. The third stage uses a message access agent.

Currently two message access protocols are available: Post Office Protocol, version 3 (POP3) and Internet Mail Access Protocol, version 4 (IMAP4). Figure 26.19 shows the position of these two protocols in the most common situation (fourth scenario).



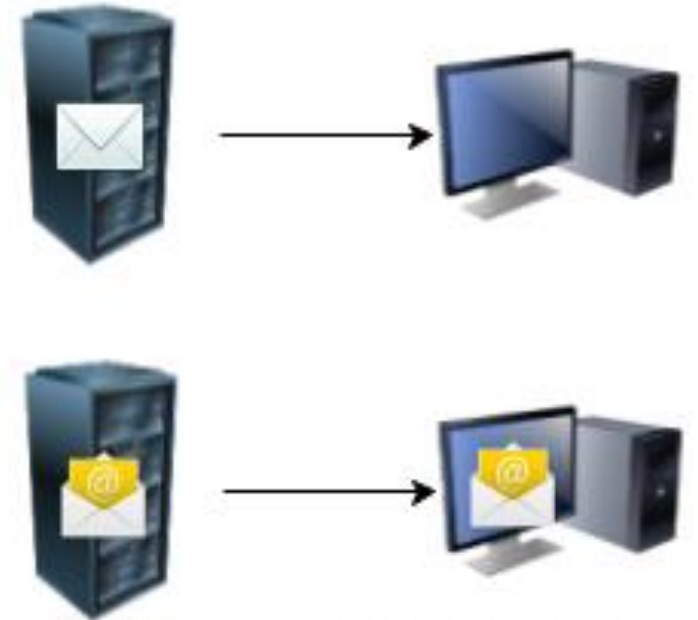
# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Serveur POP



Le mail est déplacé sur l'ordinateur.  
(il est donc effacé du serveur).

Serveur IMAP



Le mail est affiché sur l'ordinateur.  
(C'est une vue du contenu du serveur).  
Il reste sur le serveur  
(jusqu'à ce que l'utilisateur l'efface)

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## File Transfer Protocol (FTP)

**File Transfer Protocol (FTP)** is the standard mechanism provided by TCP/IP for copying a file from one host to another. Although transferring files from one system to another seems simple and straightforward, some problems must be dealt with first. For example, two systems may use different file name conventions. Two systems may have different ways to represent text and data. Two systems may have different directory structures. All these problems have been solved by FTP in a very simple and elegant approach.

FTP differs from other client/server applications in that it establishes two connections between the hosts. One connection is used for data transfer, the other for control information (commands and responses). Separation of commands and data transfer makes FTP more efficient. The control connection uses very simple rules of communication. We need to transfer only a line of command or a line of response at a time. The data connection, on the other hand, needs more complex rules due to the variety of data types transferred. However, the difference in complexity is at the FTP level, not TCP. For TCP, both connections are treated the same.

FTP uses two well-known TCP ports: Port 21 is used for the control connection, and port 20 is used for the data connection.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

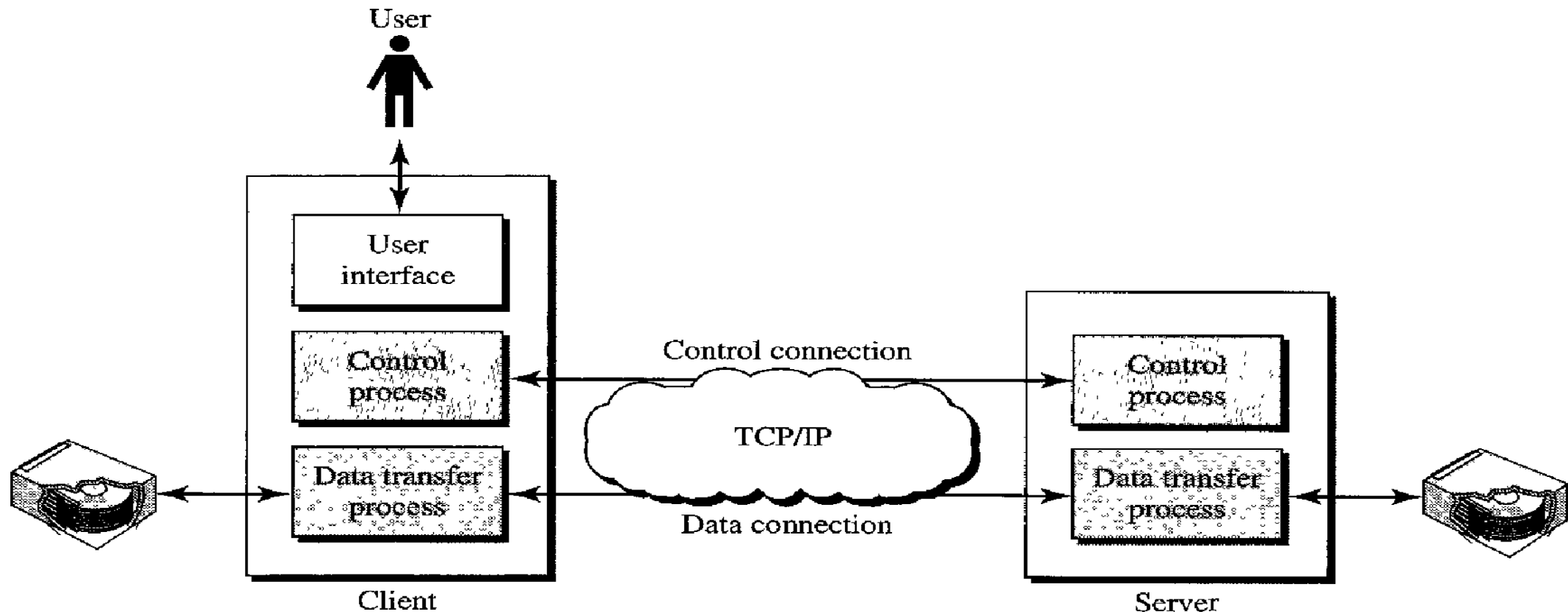
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**FTP uses the services of TCP. It needs two TCP connections.  
The well-known port 21 is used for the control connection  
and the well-known port 20 for the data connection.**

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 26.21 *FTP*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## *WWW and HTTP*

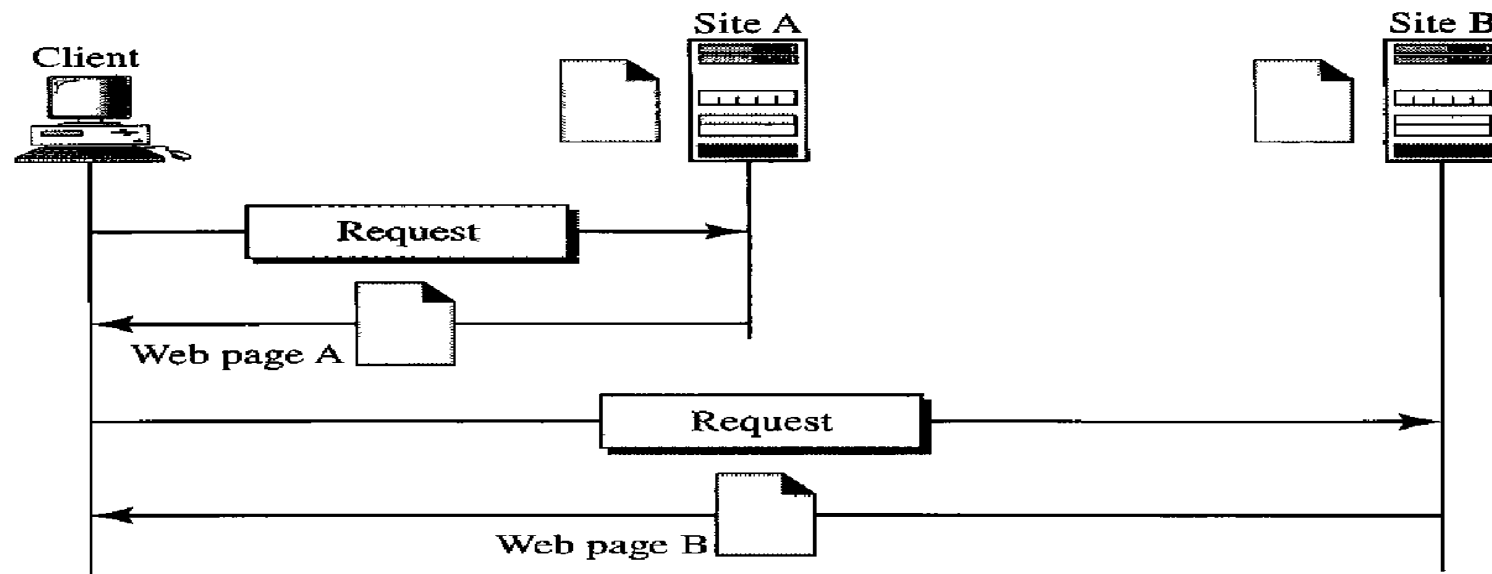
The **World Wide Web** (WWW) is a repository of information linked together from points all over the world. The WWW has a unique combination of flexibility, portability, and user-friendly features that distinguish it from other services provided by the Internet. The WWW project was initiated by CERN (European Laboratory for Particle Physics) to create a system to handle distributed resources necessary for scientific research. In this chapter we first discuss issues related to the Web. We then discuss a protocol, HTTP, that is used to retrieve information from the Web.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## ARCHITECTURE

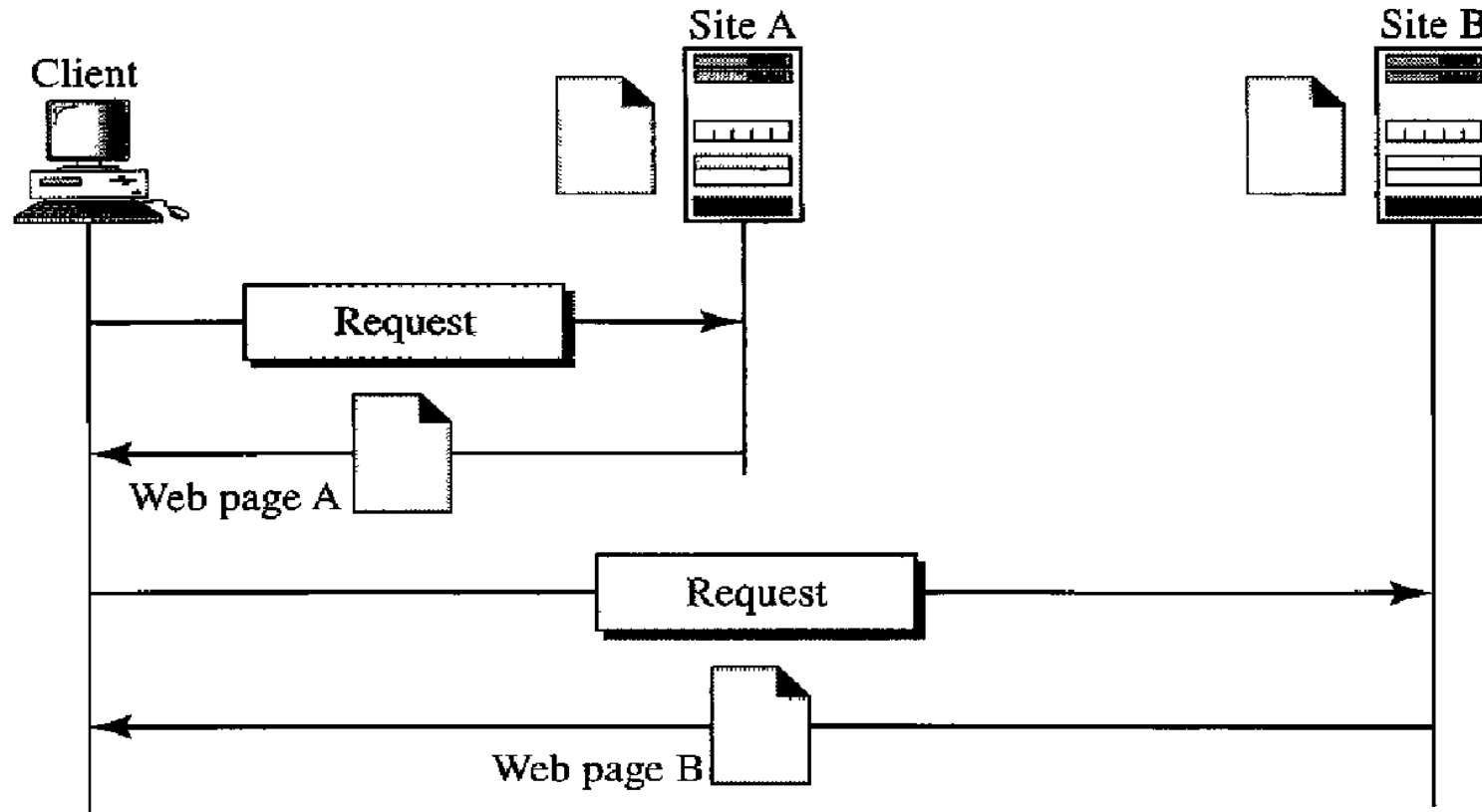
The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called *sites*, as shown in Figure 27.1.

**Figure 27.1** *Architecture of WWW*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

**Figure 27.1** *Architecture of WWW*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

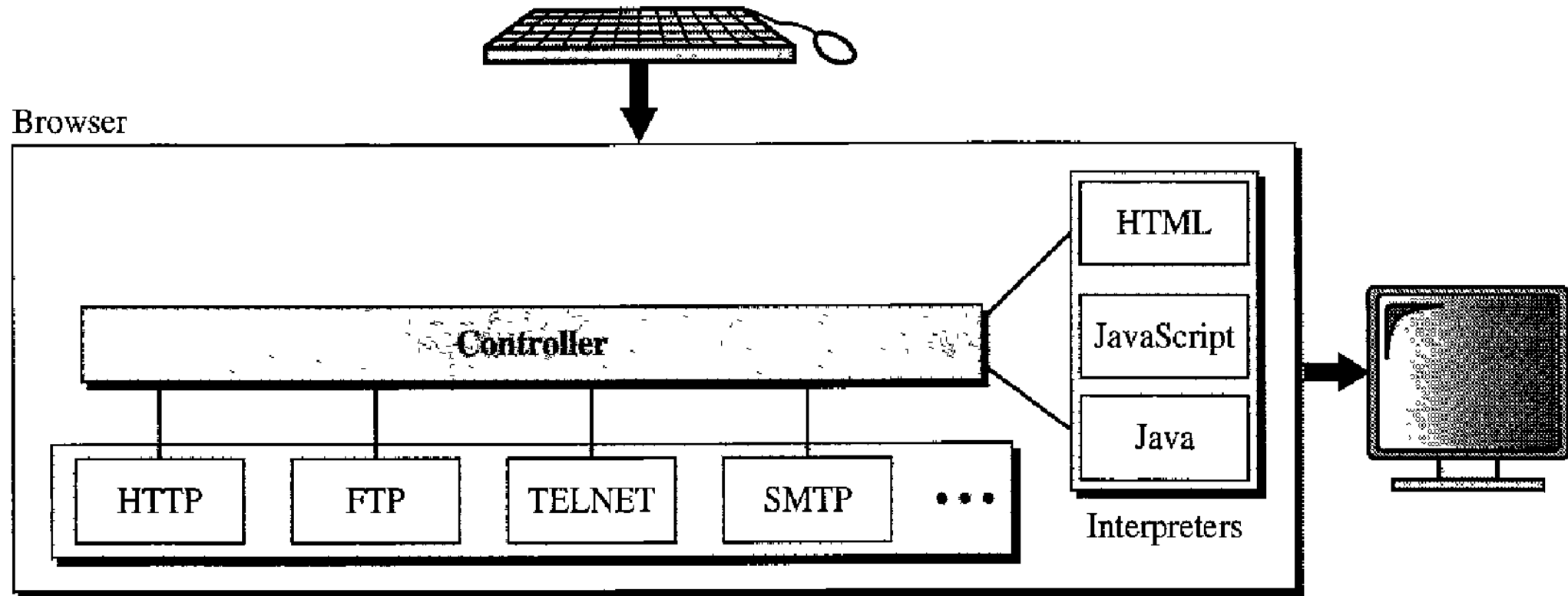
## Client (Browser)

A variety of vendors offer commercial browsers that interpret and display a Web document, and all use nearly the same architecture. Each **browser** usually consists of three parts: a controller, client protocol, and interpreters. The controller receives input from the keyboard or the mouse and uses the client programs to access the document. After the document has been accessed, the controller uses one of the interpreters to display the document on the screen. The client protocol can be one of the protocols described previously such as FTP or HTTP (described later in the chapter). The interpreter can be HTML, Java, or JavaScript, depending on the type of document. We discuss the use of these interpreters based on the document type later in the chapter (see Figure 27.2).



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 27.2 *Browser*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## Server

The Web page is stored at the server. Each time a client request arrives, the corresponding document is sent to the client. To improve efficiency, servers normally store requested files in a cache in memory; memory is faster to access than disk. A server can also become more efficient through multithreading or multiprocessing. In this case, a server can answer more than one request at a time.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## Uniform Resource Locator

A client that wants to access a Web page needs the address. To facilitate the access of documents distributed throughout the world, HTTP uses locators. The **uniform resource locator (URL)** is a standard for specifying any kind of information on the Internet. The URL defines four things: protocol, host computer, port, and path (see Figure 27.3).

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**Figure 27.3** *URL*

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

The *protocol* is the client/server program used to retrieve the document. Many different protocols can retrieve a document; among them are FTP or HTTP. The most common today is HTTP.

The **host** is the computer on which the information is located, although the name of the computer can be an alias. Web pages are usually stored in computers, and computers are given alias names that usually begin with the characters “www”. This is not mandatory, however, as the host can be any name given to the computer that hosts the Web page.

The URL can optionally contain the port number of the server. If the *port* is included, it is inserted between the host and the path, and it is separated from the host by a colon.

**Path** is the pathname of the file where the information is located. Note that the path can itself contain slashes that, in the UNIX operating system, separate the directories from the subdirectories and files.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## 27.2 WEB DOCUMENTS

The documents in the WWW can be grouped into three broad categories: static, dynamic, and active. The category is based on the time at which the contents of the document are determined.

### Static Documents

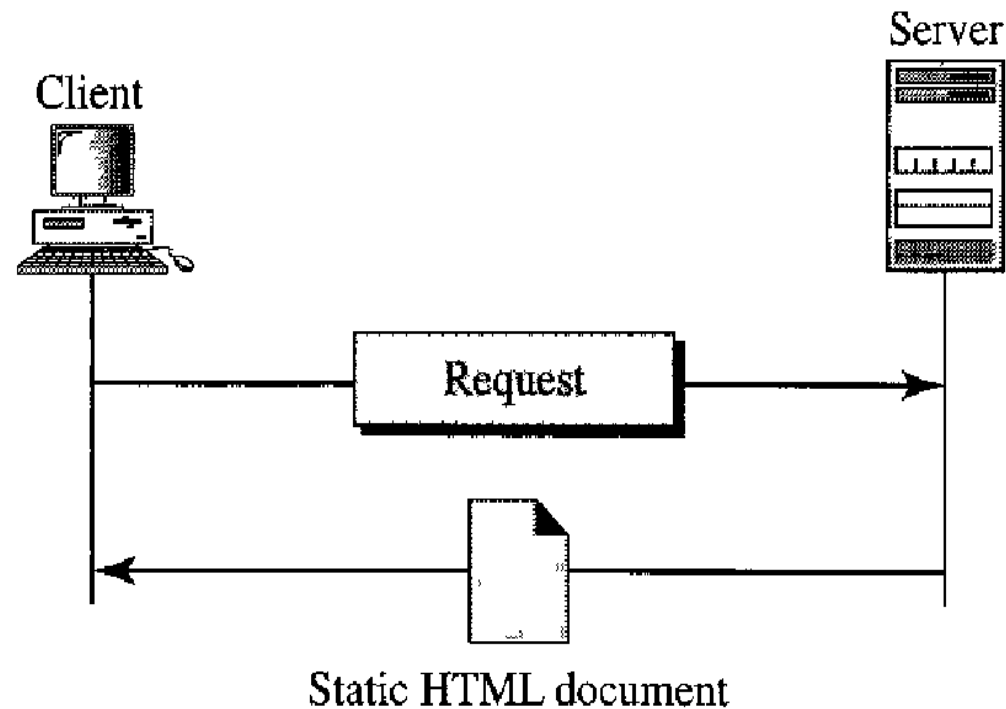
**Static documents** are fixed-content documents that are created and stored in a server. The client can get only a copy of the document. In other words, the contents of the file are determined when the file is created, not when it is used. Of course, the contents in the server can be changed, but the user cannot change them. When a client accesses the document, a copy of the document is sent. The user can then use a browsing program to display the document (see Figure 27.4).

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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**Figure 27.4** *Static document*

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## *HTML*

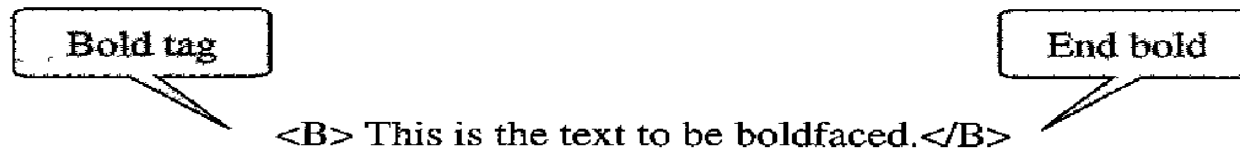
**Hypertext Markup Language (HTML)** is a language for creating Web pages. The term *markup language* comes from the book publishing industry. Before a book is typeset and printed, a copy editor reads the manuscript and puts marks on it. These marks tell the compositor how to format the text. For example, if the copy editor wants part of a line to be printed in boldface, he or she draws a wavy line under that part. In the same way, data for a Web page are formatted for interpretation by a browser.

Let us clarify the idea with an example. To make part of a text displayed in boldface with HTML, we put beginning and ending boldface **tags** (marks) in the text, as shown in Figure 27.5.

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**Figure 27.5** *Boldface tags*

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

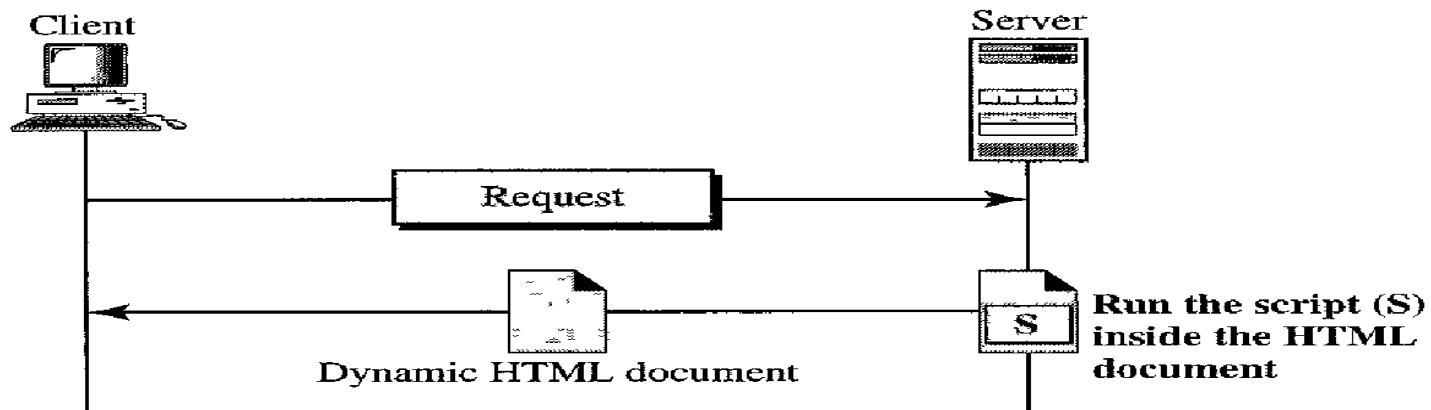
## Dynamic Documents

A **dynamic document** is created by a Web server whenever a browser requests the document. When a request arrives, the Web server runs an application program or a script that creates the dynamic document. The server returns the output of the program or script as a response to the browser that requested the document. Because a fresh document is created for each request, the contents of a dynamic document can vary from one request to another. A very simple example of a dynamic document is the retrieval of the time and date from a server. Time and date are kinds of information that are dynamic in that they change from moment to moment. The client can ask the server to run a program such as the *date* program in UNIX and send the result of the program to the client.



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

**Figure 27.9** *Dynamic document using server-site script*

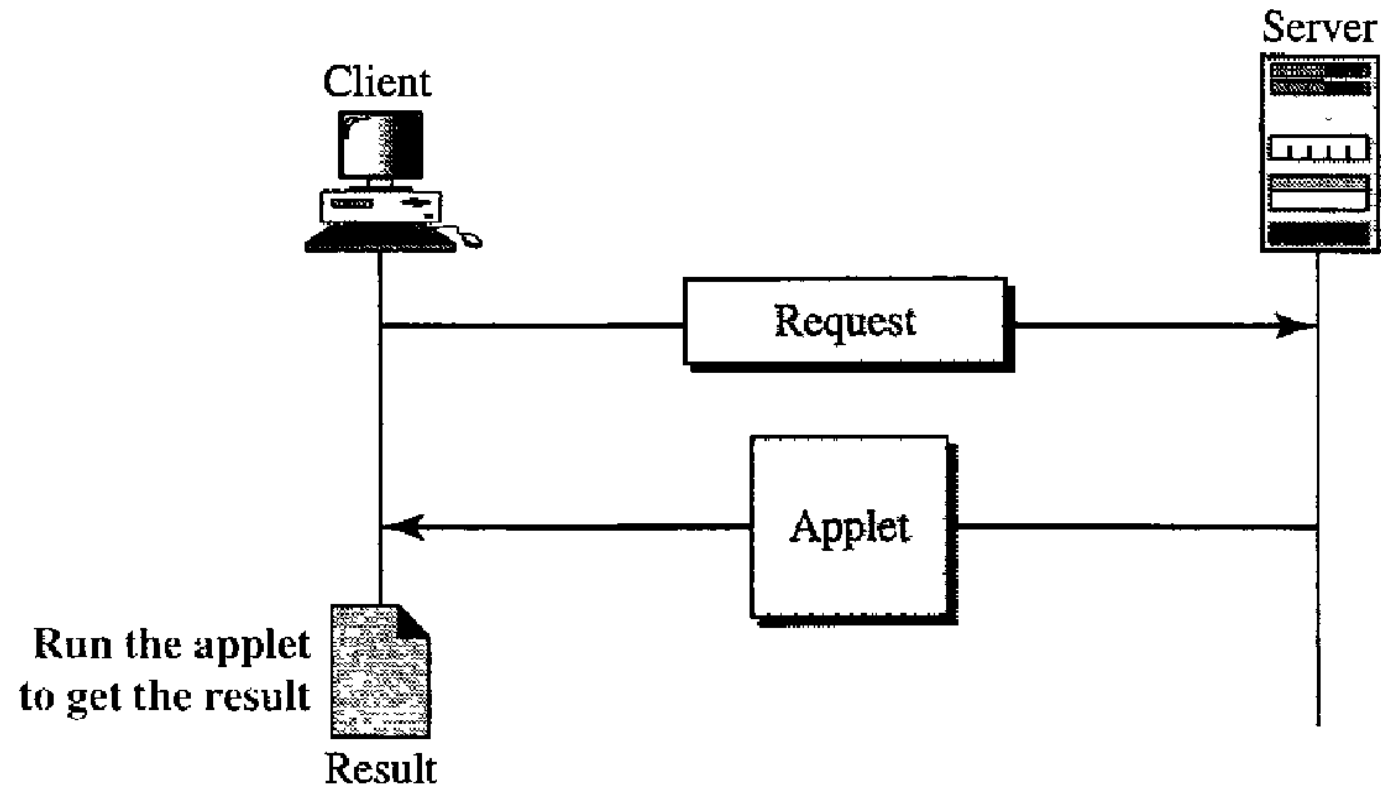


A few technologies have been involved in creating dynamic documents using scripts. Among the most common are **Hypertext Preprocessor (PHP)**, which uses the Perl language; **Java Server Pages (JSP)**, which uses the Java language for scripting; **Active Server Pages (ASP)**, a Microsoft product which uses Visual Basic language for scripting; and **ColdFusion**, which embeds SQL database queries in the HTML document.

**Dynamic documents are sometimes referred to as server-site dynamic documents.**

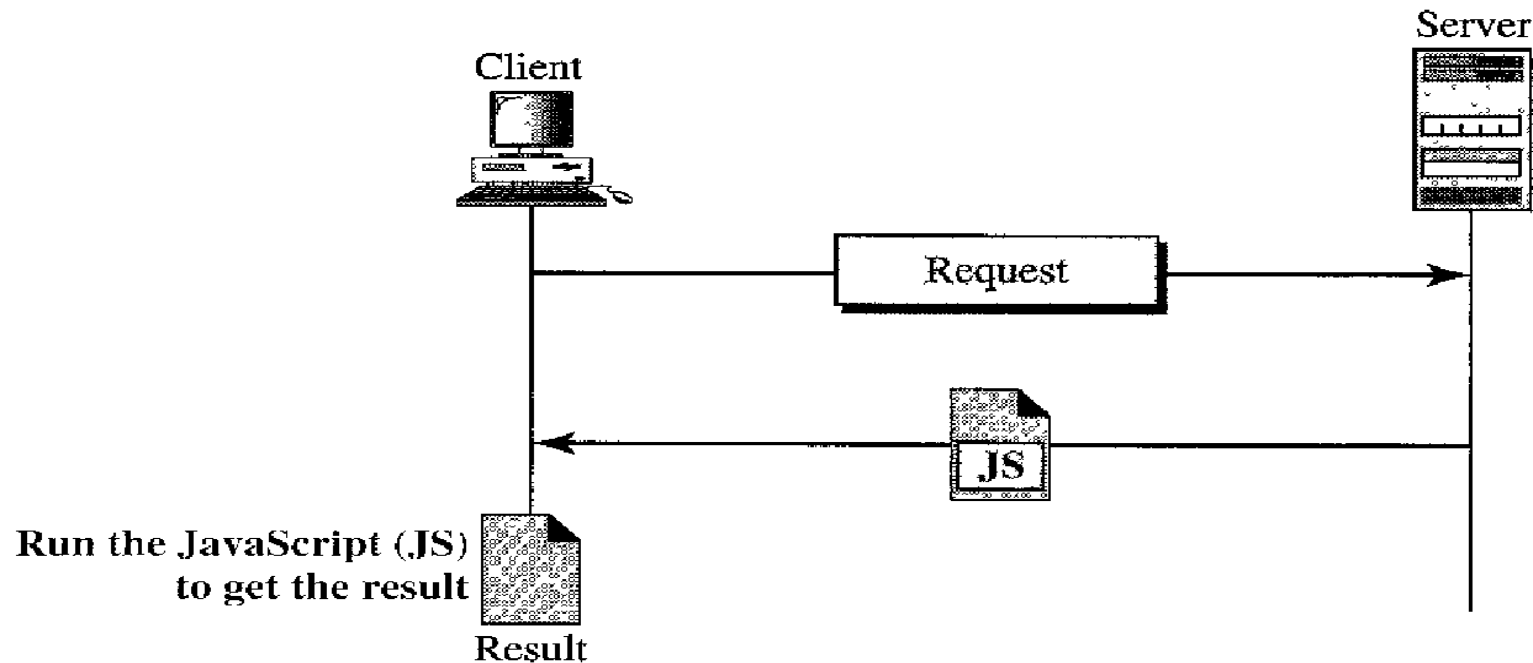
# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 27.10 *Active document using Java applet*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 27.11 *Active document using client-site script*



**Active documents are sometimes referred to as client-site dynamic documents.**

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## 27.3 HTTP

The **Hypertext Transfer Protocol (HTTP)** is a protocol used mainly to access data on the World Wide Web. HTTP functions as a combination of FTP and SMTP. It is similar to FTP because it transfers files and uses the services of TCP. However, it is much simpler than FTP because it uses only one TCP connection. There is no separate control connection; only data are transferred between the client and the server.

HTTP is like SMTP because the data transferred between the client and the server look like SMTP messages. In addition, the format of the messages is controlled by MIME-like headers. Unlike SMTP, the HTTP messages are not destined to be read by humans; they are read and interpreted by the HTTP server and HTTP client (browser). SMTP messages are stored and forwarded, but HTTP messages are delivered immediately. The commands from the client to the server are embedded in a request message. The contents of the requested file or other information are embedded in a response message. HTTP uses the services of TCP on well-known port 80.

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**HTTP uses the services of TCP on well-known port 80.**

---

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## *Network Management: SNMP*

We can define **network management** as monitoring, testing, configuring, and troubleshooting network components to meet a set of requirements defined by an organization. These requirements include the smooth, efficient operation of the network that provides the predefined quality of service for users. To accomplish this task, a network management system uses hardware, software, and humans. In this chapter, first we briefly discuss the functions of a network management system. Then we concentrate on the most common management system, the Simple Network Management Protocol (SNMP).

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## NETWORK MANAGEMENT SYSTEM

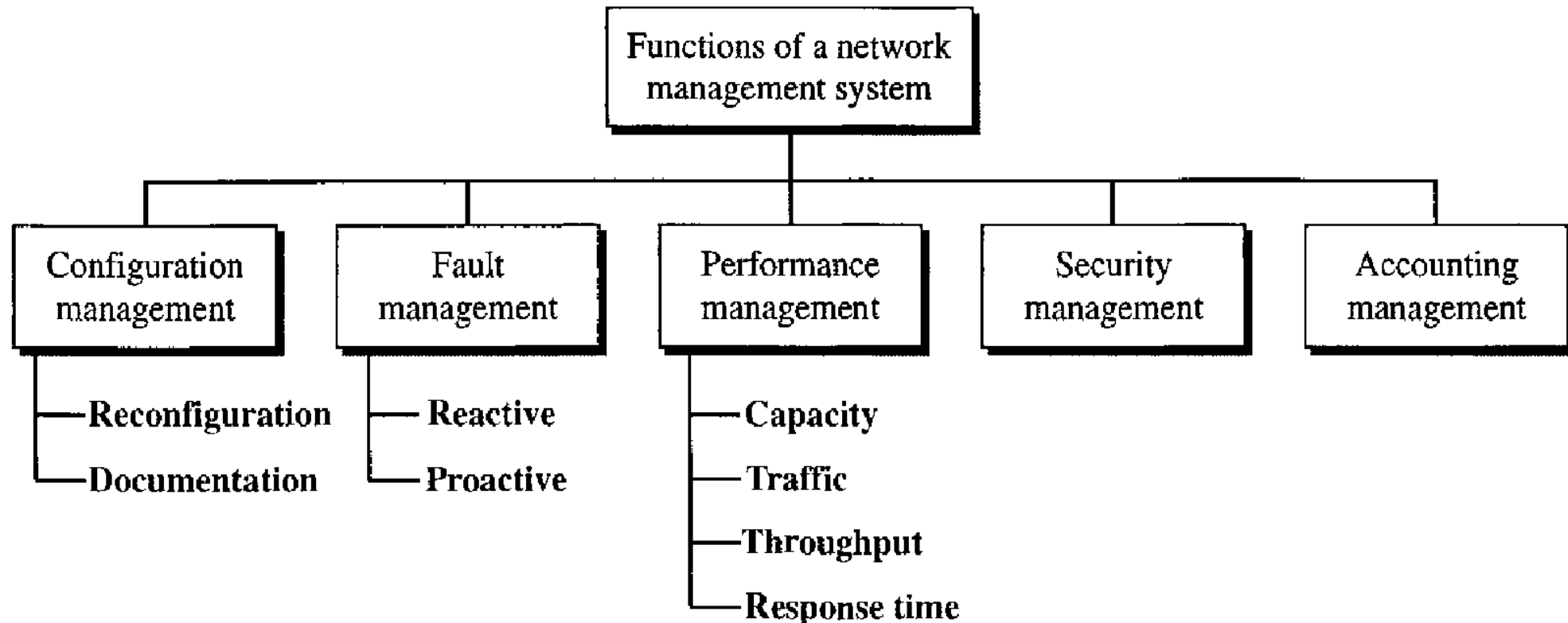
We can say that the functions performed by a network management system can be divided into five broad categories: configuration management, fault management, performance management, security management, and accounting management, as shown in Figure 28.1.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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**Figure 28.1** *Functions of a network management system*

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**Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP**

## **SIMPLE NETWORK MANAGEMENT PROTOCOL (SNMP)**

The **Simple Network Management Protocol (SNMP)** is a framework for managing devices in an internet using the TCP/IP protocol suite. It provides a set of fundamental operations for monitoring and maintaining an internet.

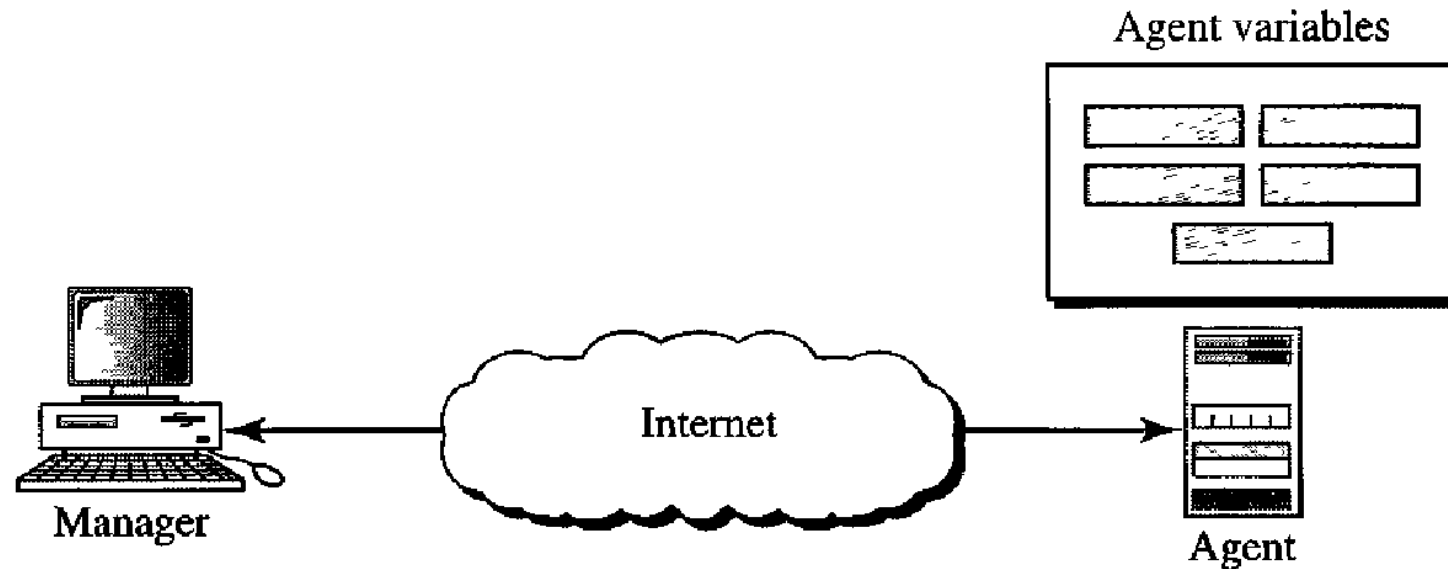
### **Concept**

SNMP uses the concept of manager and agent. That is, a manager, usually a host, controls and monitors a set of agents, usually routers (see Figure 28.2).



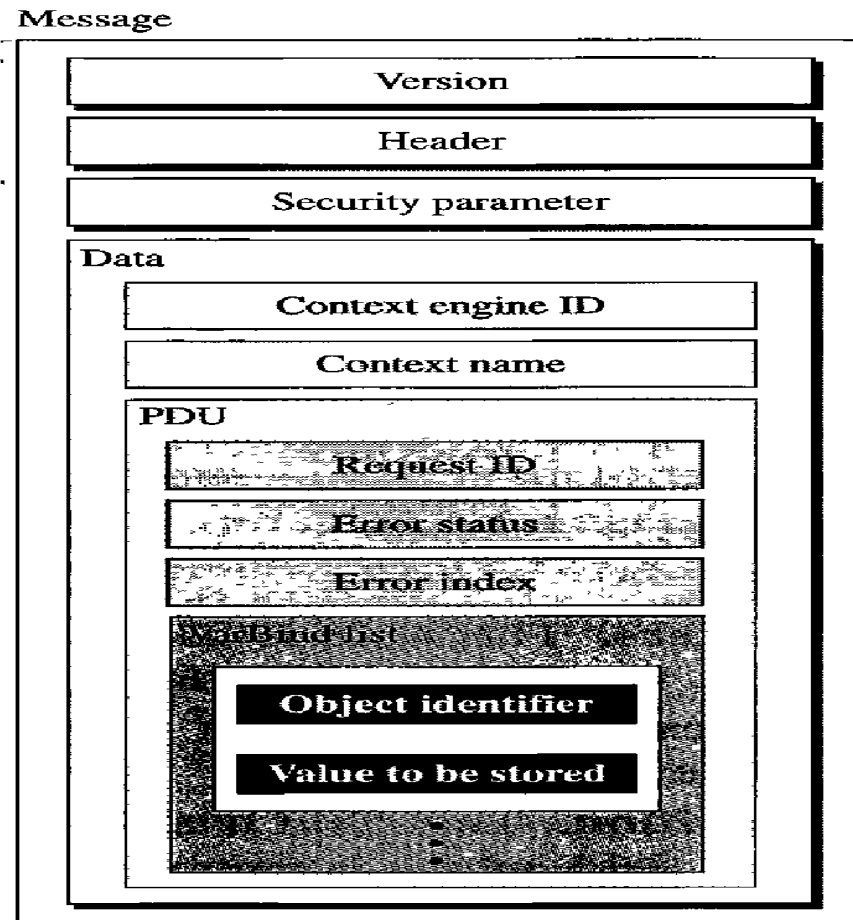
# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 28.2 *SNMP concept*



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**Figure 28.22** *SNMP message*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## *Multimedia*

Recent advances in technology have changed our use of audio and video. In the past, we listened to an audio broadcast through a radio and watched a video program broadcast through a TV. We used the telephone network to interactively communicate with another party. But times have changed. People want to use the Internet not only for text and image communications, but also for audio and video services. In this chapter, we concentrate on applications that use the Internet for audio and video services.

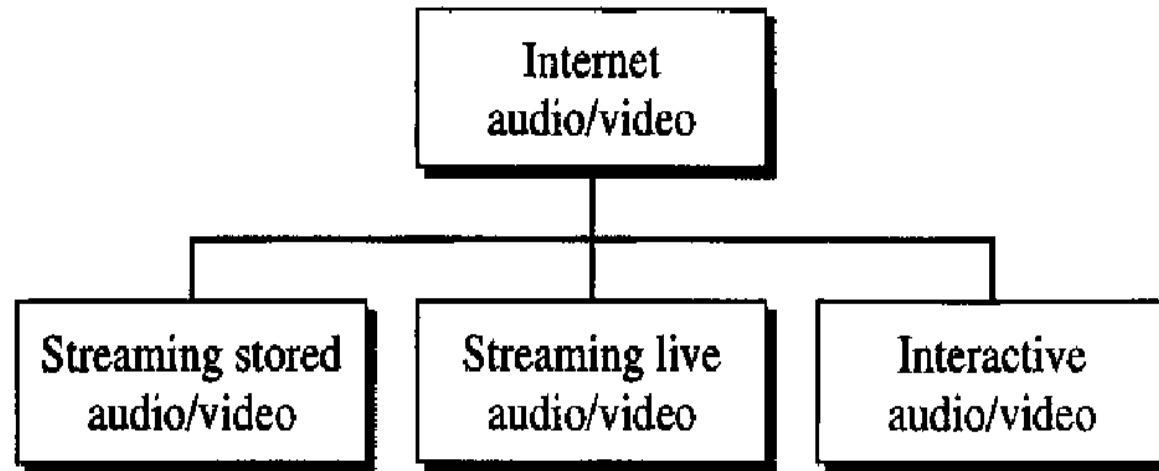
We can divide audio and video services into three broad categories: **streaming stored audio/video**, **streaming live audio/video**, and **interactive audio/video**, as shown in Figure 29.1. Streaming means a user can listen to (or watch) the file after the downloading has started.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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Figure 29.1 *Internet audio/video*

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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**Streaming stored audio/video refers to on-demand requests for compressed audio/video files.**

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**Streaming live audio/video refers to the broadcasting of radio and TV programs through the Internet.**

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**Interactive audio/video refers to the use of the Internet for interactive audio/video applications.**

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## Digitizing Audio

When sound is fed into a microphone, an electronic analog signal is generated which represents the sound amplitude as a function of time. The signal is called an *analog audio signal*. An analog signal, such as audio, can be digitized to produce a digital signal. According to the Nyquist theorem, if the highest frequency of the signal is  $f$ , we need to sample the signal  $2f$  times per second. There are other methods for digitizing an audio signal, but the principle is the same.

Voice is sampled at 8000 samples per second with 8 bits per sample. This results in a digital signal of 64 kbps. Music is sampled at 44,100 samples per second with 16 bits per sample. This results in a digital signal of 705.6 kbps for monaural and 1.411 Mbps for stereo.

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## Digitizing Video

A video consists of a sequence of frames. If the frames are displayed on the screen fast enough, we get an impression of motion. The reason is that our eyes cannot distinguish the rapidly flashing frames as individual ones. There is no standard number of frames per second; in North America 25 frames per second is common. However, to avoid a

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condition known as flickering, a frame needs to be refreshed. The TV industry repaints each frame twice. This means 50 frames need to be sent, or if there is memory at the sender site, 25 frames with each frame repainted from the memory.

Each frame is divided into small grids, called picture elements or **pixels**. For black-and-white TV, each 8-bit pixel represents one of 256 different gray levels. For a color TV, each pixel is 24 bits, with 8 bits for each primary color (red, green, and blue).

We can calculate the number of bits in 1 s for a specific resolution. In the lowest resolution a color frame is made of  $1024 \times 768$  pixels. This means that we need

$$2 \times 25 \times 1024 \times 768 \times 24 = 944 \text{ Mbps}$$

This data rate needs a very high data rate technology such as SONET. To send video using lower-rate technologies, we need to compress the video.

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***Compression is needed to send video over the Internet.***

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## 29.2 AUDIO AND VIDEO COMPRESSION

To send audio or video over the Internet requires **compression**. In this section, we discuss audio compression first and then video compression.

### Audio Compression

Audio compression can be used for speech or music. For speech, we need to compress a 64-kHz digitized signal; for music, we need to compress a 1.411-MHz signal. Two categories of techniques are used for audio compression: predictive encoding and perceptual encoding.

#### *Predictive Encoding*

In **predictive encoding**, the differences between the samples are encoded instead of encoding all the sampled values. This type of compression is normally used for speech. Several standards have been defined such as GSM (13 kbps), G.729 (8 kbps), and G.723.3 (6.4 or 5.3 kbps). Detailed discussions of these techniques are beyond the scope of this book.

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## *Perceptual Encoding: MP3*

The most common compression technique that is used to create CD-quality audio is based on the **perceptual encoding** technique. As we mentioned before, this type of audio needs at least 1.411 Mbps; this cannot be sent over the Internet without compression. **MP3** (MPEG audio layer 3), a part of the MPEG standard (discussed in the video compression section), uses this technique.

Perceptual encoding is based on the science of psychoacoustics, which is the study of how people perceive sound. The idea is based on flaws in our auditory system: Some sounds can mask other sounds. Masking can happen in frequency and time. In **frequency masking**, a loud sound in a frequency range can partially or totally mask a softer sound in another frequency range. For example, we cannot hear what our dance partner says in a room where a loud heavy metal band is performing. In **temporal masking**, a loud sound can numb our ears for a short time even after the sound has stopped.

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MP3 uses these two phenomena, frequency and temporal masking, to compress audio signals. The technique analyzes and divides the spectrum into several groups. Zero bits are allocated to the frequency ranges that are totally masked. A small number of bits are allocated to the frequency ranges that are partially masked. A larger number of bits are allocated to the frequency ranges that are not masked.

MP3 produces three data rates: 96 kbps, 128 kbps, and 160 kbps. The rate is based on the range of the frequencies in the original analog audio.

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## Video Compression

As we mentioned before, video is composed of multiple frames. Each frame is one image. We can compress video by first compressing images. Two standards are prevalent in the market. **Joint Photographic Experts Group (JPEG)** is used to compress images. **Moving Picture Experts Group (MPEG)** is used to compress video. We briefly discuss JPEG and then MPEG.

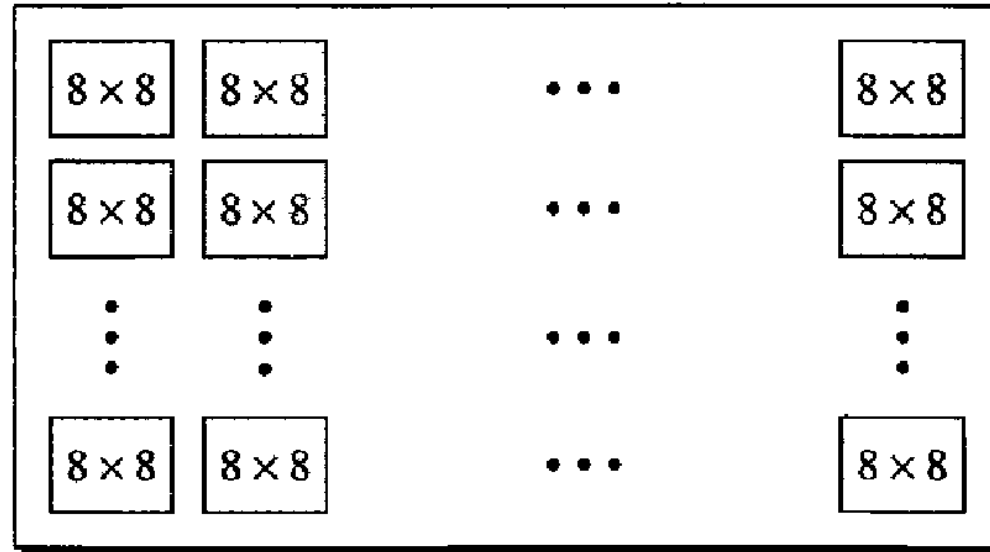
### *Image Compression: JPEG*

As we discussed previously, if the picture is not in color (gray scale), each pixel can be represented by an 8-bit integer (256 levels). If the picture is in color, each pixel can be represented by 24 bits ( $3 \times 8$  bits), with each 8 bits representing red, blue, or green (RBG). To simplify the discussion, we concentrate on a gray scale picture.

In JPEG, a gray scale picture is divided into blocks of  $8 \times 8$  pixels (see Figure 29.2).

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**Figure 29.2** *JPEG gray scale*

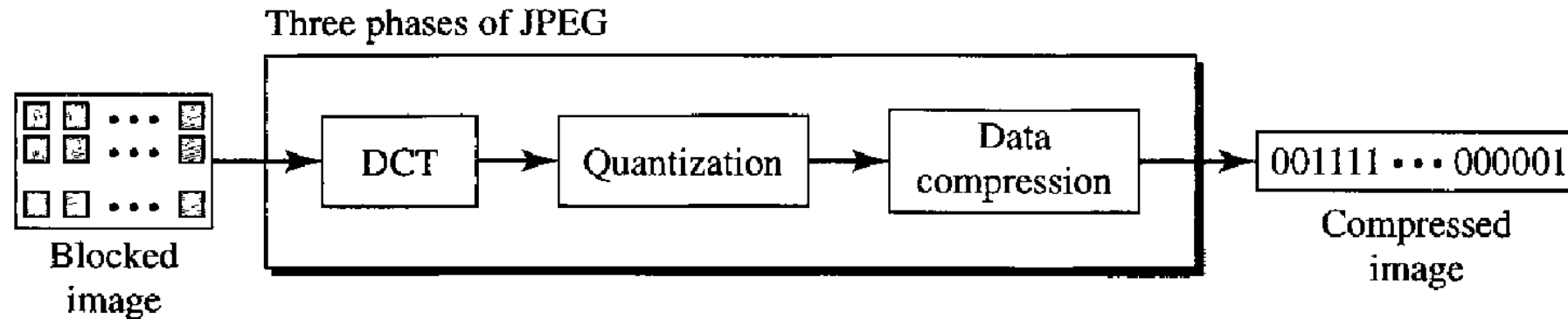


# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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**Figure 29.3** *JPEG process*

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**Discrete Cosine Transform (DCT)** In this step, each block of 64 pixels goes through a transformation called the **discrete cosine transform (DCT)**. The transformation changes the 64 values so that the relative relationships between pixels are kept but the redundancies are revealed. We do not give the formula here, but we do show the results of the transformation for three cases.

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## *Video Compression: MPEG*

The Moving Picture Experts Group method is used to compress video. In principle, a motion picture is a rapid flow of a set of frames, where each frame is an image. In other words, a frame is a spatial combination of pixels, and a video is a temporal combination of frames that are sent one after another. Compressing video, then, means spatially compressing each frame and temporally compressing a set of frames.

**Spatial Compression** The **spatial compression** of each frame is done with JPEG (or a modification of it). Each frame is a picture that can be independently compressed.

**Temporal Compression** In **temporal compression**, redundant frames are removed. When we watch television, we receive 50 frames per second. However, most of the consecutive frames are almost the same. For example, when someone is talking, most of the frame is the same as the previous one except for the segment of the frame around the lips, which changes from one frame to another.

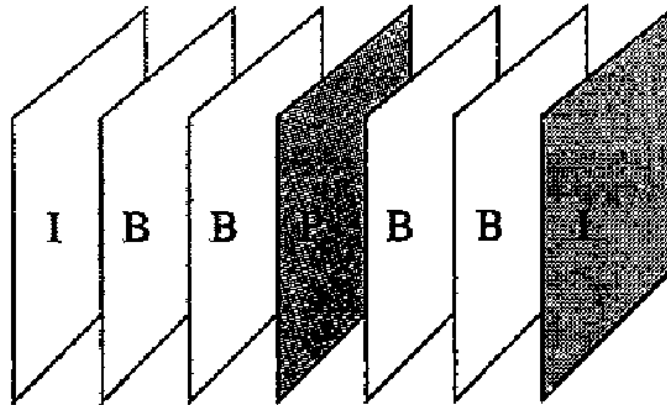
To temporally compress data, the MPEG method first divides frames into three categories: I-frames, P-frames, and B-frames.

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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**Figure 29.8** *MPEG frames*

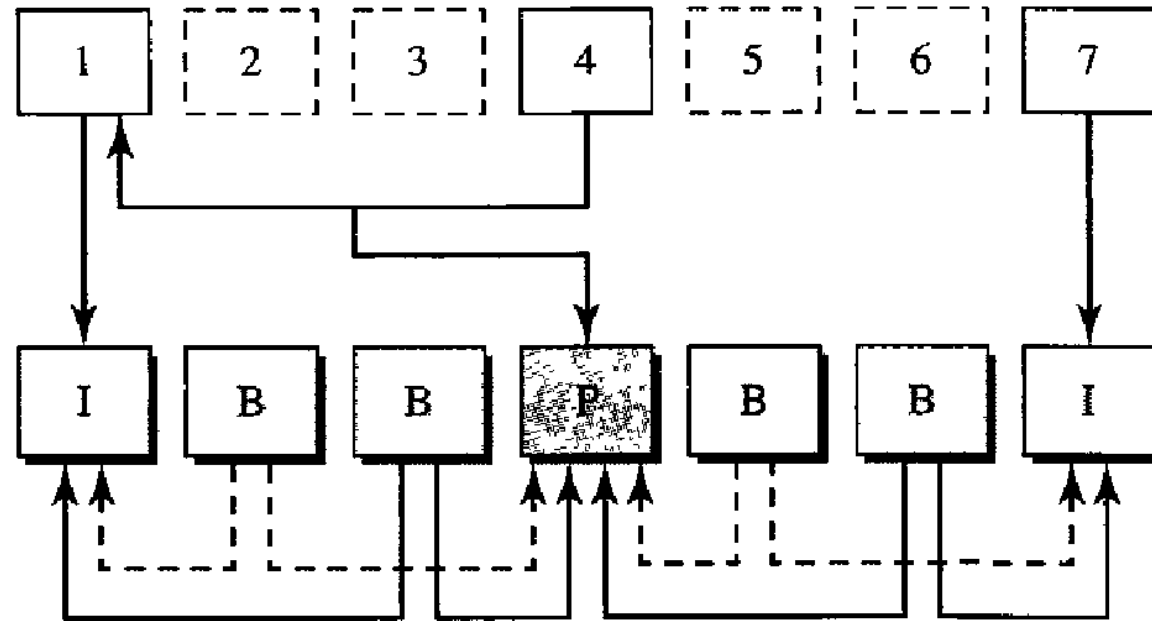
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**Figure 29.9** *MPEG frame construction*



MPEG has gone through two versions. MPEG1 was designed for a CD-ROM with a data rate of 1.5 Mbps. MPEG2 was designed for high-quality DVD with a data rate of 3 to 6 Mbps.

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## **STREAMING STORED AUDIO/VIDEO**

Now that we have discussed digitizing and compressing audio/video, we turn our attention to specific applications. The first is streaming stored audio and video. Down-types of files. To understand the concept, let us discuss four approaches, each with a different complexity.

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## **STREAMING LIVE AUDIO/VIDEO**

Streaming live audio/video is similar to the broadcasting of audio and video by radio and TV stations. Instead of broadcasting to the air, the stations broadcast through the Internet. There are several similarities between streaming stored audio/video and streaming live audio/video. They are both sensitive to delay; neither can accept retransmission. However, there is a difference. In the first application, the communication is unicast and on-demand. In the second, the communication is multicast and live. Live streaming is better suited to the multicast services of IP and the use of protocols such as UDP and RTP (discussed later). However, presently, live streaming is still using TCP and multiple unicasting instead of multicasting. There is still much progress to be made in this area.

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## **REAL-TIME INTERACTIVE AUDIO/VIDEO**

In real-time interactive audio/video, people communicate with one another in real time. The Internet phone or voice over IP is an example of this type of application. Video conferencing is another example that allows people to communicate visually and orally.

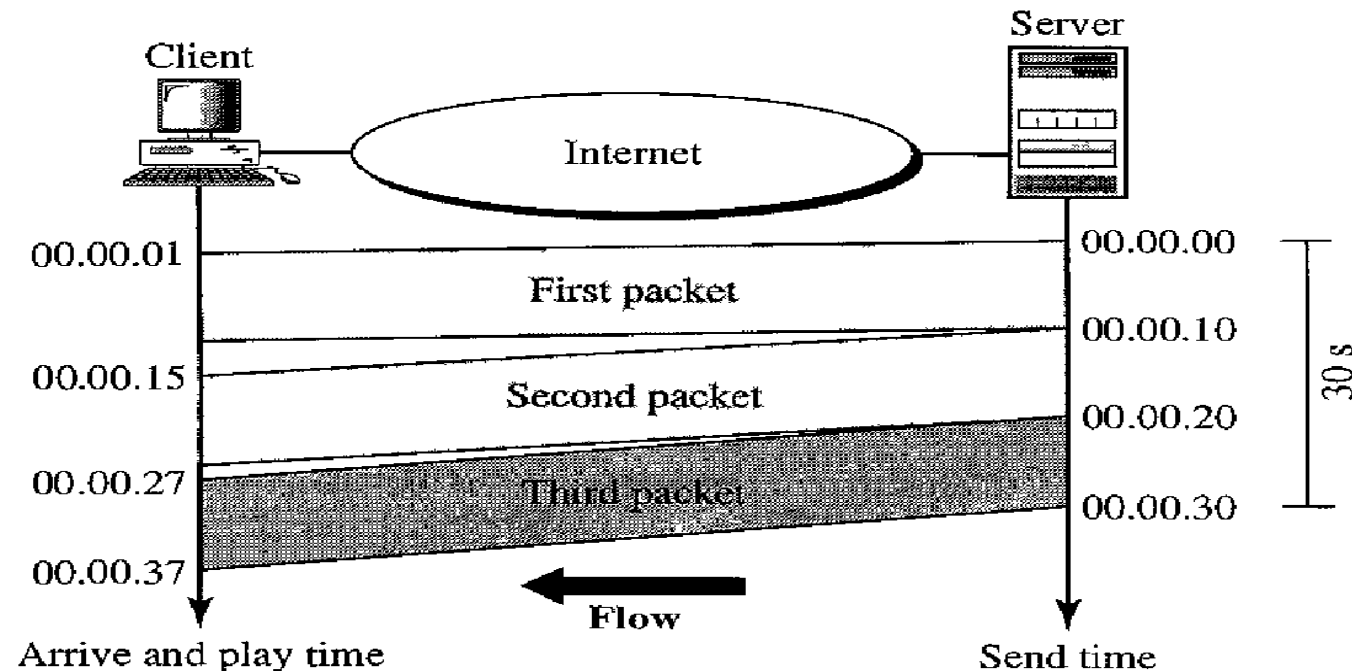
### **Characteristics**

Before addressing the protocols used in this class of applications, we discuss some characteristics of real-time audio/video communication.

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**Jitter is introduced in real-time data by the delay between packets.**

**Figure 29.15** *Jitter*



# **Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP**

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**Translation means changing the encoding of a payload to a lower quality to match the bandwidth of the receiving network.**

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**Mixing means combining several streams of traffic into one stream.**

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**TCP, with all its sophistication, is not suitable for interactive multimedia traffic because we cannot allow retransmission of packets.**

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**UDP is more suitable than TCP for interactive traffic. However, we need the services of RTP, another transport layer protocol, to make up for the deficiencies of UDP.**

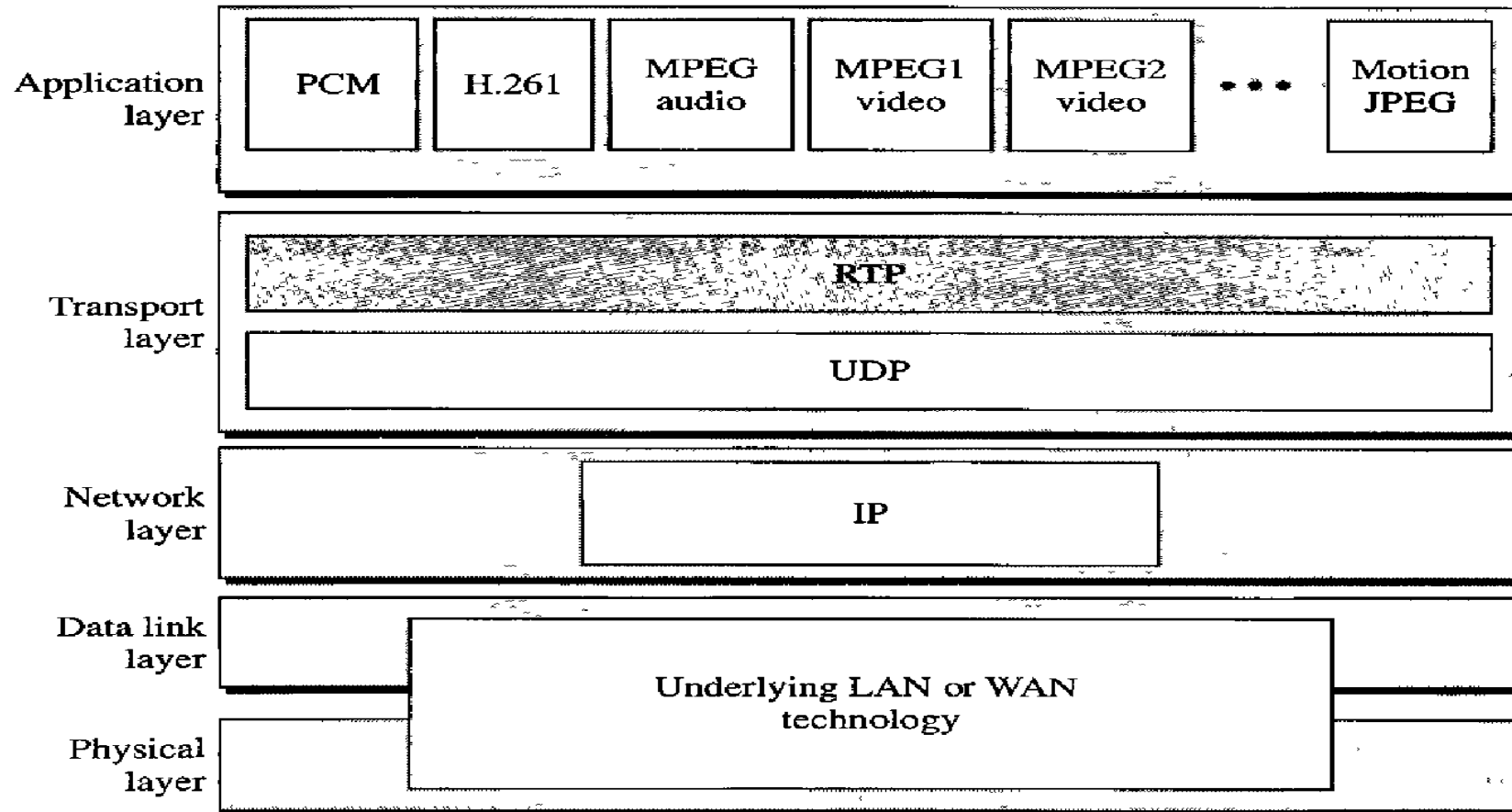
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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

**Real-time Transport Protocol (RTP)** is the protocol designed to handle real-time traffic on the Internet. RTP does not have a delivery mechanism (multicasting, port numbers, and so on); it must be used with UDP. RTP stands between UDP and the application program. The main contributions of RTP are time-stamping, sequencing, and mixing facilities. Figure 29.18 shows the position of RTP in the protocol suite.

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**Figure 29.18** *RTP*





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**Table 29.1** *Payload types*

| <i>Type</i> | <i>Application</i> | <i>Type</i> | <i>Application</i> | <i>Type</i> | <i>Application</i> |
|-------------|--------------------|-------------|--------------------|-------------|--------------------|
| 0           | PCMu Audio         | 7           | LPC audio          | 15          | G728 audio         |
| 1           | 1016               | 8           | PCMA audio         | 26          | Motion JPEG        |
| 2           | G721 audio         | 9           | G722 audio         | 31          | H.261              |
| 3           | GSM audio          | 10–11       | L16 audio          | 32          | MPEG1 video        |
| 5–6         | DV14 audio         | 14          | MPEG audio         | 33          | MPEG2 video        |

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## UDP Port

Although RTP is itself a transport layer protocol, the RTP packet is not encapsulated directly in an IP datagram. Instead, RTP is treated as an application program and is encapsulated in a UDP user datagram. However, unlike other application programs, no well-known port is assigned to RTP. The port can be selected on demand with only one restriction: The port number must be an even number. The next number (an odd number) is used by the companion of RTP, Real-time Transport Control Protocol (RTCP).

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**RTP uses a temporary even-numbered UDP port.**

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## RTCP

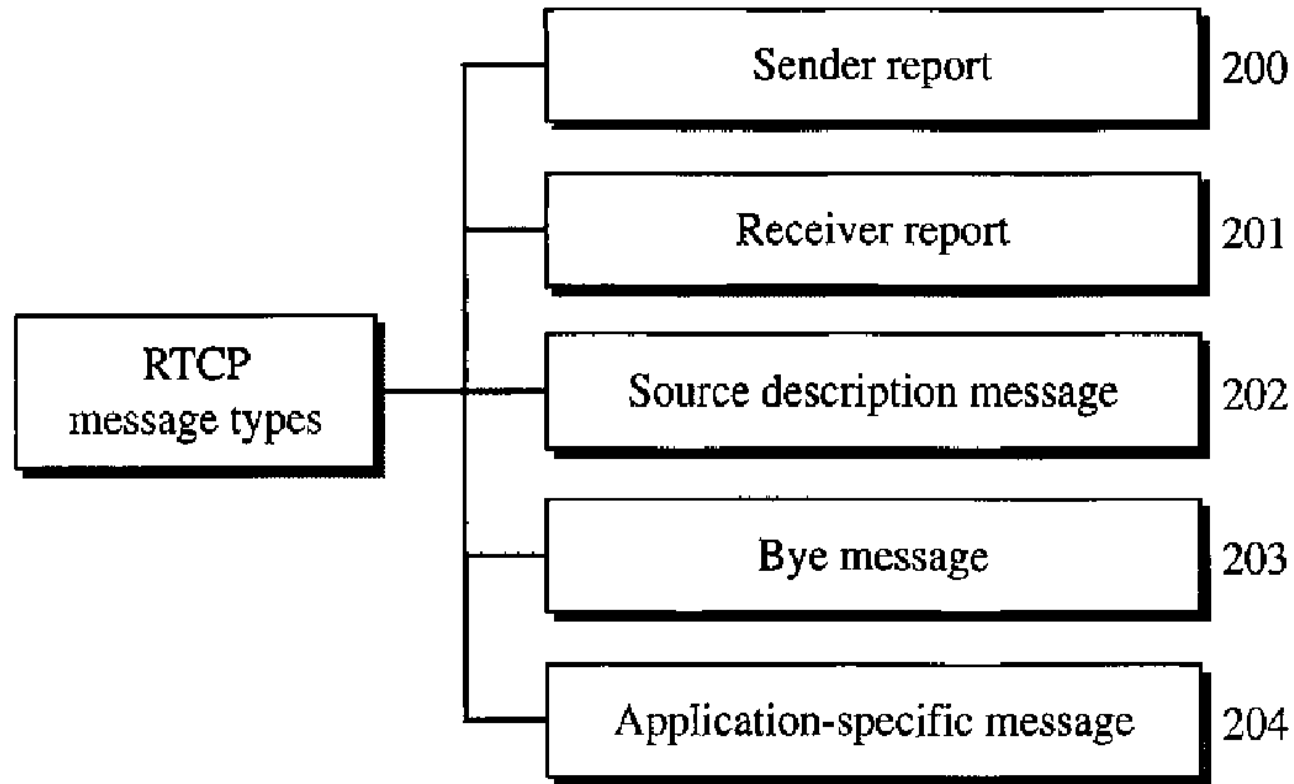
RTP allows only one type of message, one that carries data from the source to the destination. In many cases, there is a need for other messages in a session. These messages control the flow and quality of data and allow the recipient to send feedback to the source or sources. **Real-time Transport Control Protocol (RTCP)** is a protocol designed for this purpose. RTCP has five types of messages, as shown in Figure 29.20. The number next to each box defines the type of the message.

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Figure 29.20 *RTCP message types*

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## UDP Port

RTCP, like RTP, does not use a well-known UDP port. It uses a temporary port. The UDP port chosen must be the number immediately following the UDP port selected for RTP. It must be an odd-numbered port.

---

**RTCP uses an odd-numbered UDP port number that follows the port number selected for RTP.**

---

# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## VOICE OVER IP

Let us concentrate on one real-time interactive audio/video application: **voice over IP**, or Internet telephony. The idea is to use the Internet as a telephone network with some additional capabilities. Instead of communicating over a circuit-switched network, this application allows communication between two parties over the packet-switched Internet. Two protocols have been designed to handle this type of communication: SIP and H.323. We briefly discuss both.

## SIP

The **Session Initiation Protocol (SIP)** was designed by IETF. It is an application layer protocol that establishes, manages, and terminates a multimedia session (call). It can be used to create two-party, multiparty, or multicast sessions. SIP is designed to be independent of the underlying transport layer; it can run on UDP, TCP, or SCTP.

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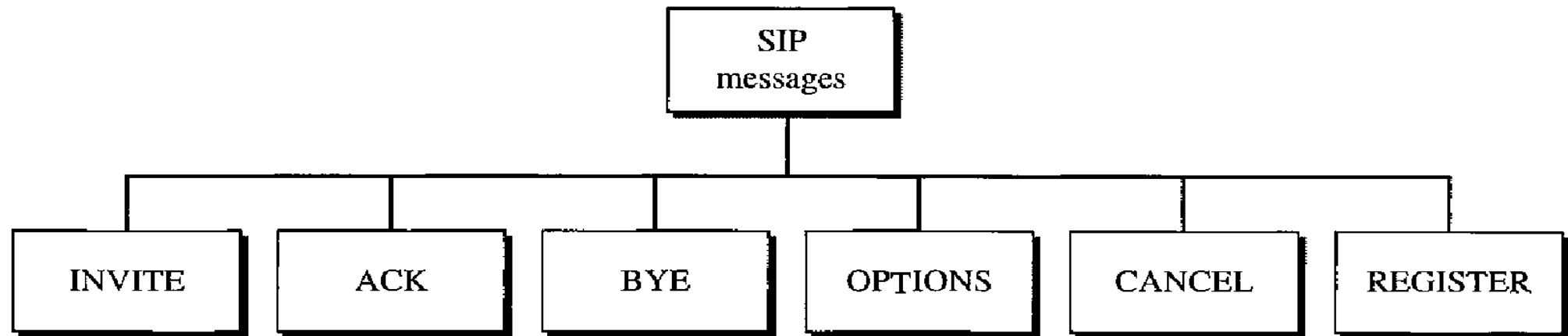
## *Messages*

SIP is a text-based protocol, as is HTTP. SIP, like HTTP, uses messages. Six messages are defined, as shown in Figure 29.21.

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**Figure 29.21**    *SIP messages*

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# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

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**Figure 29.22** *SIP formats*

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sip:bob@201.23.45.78

IPv4 address

sip:bob@fhda.edu

E-mail address

sip:bob@408-864-8900

Phone number

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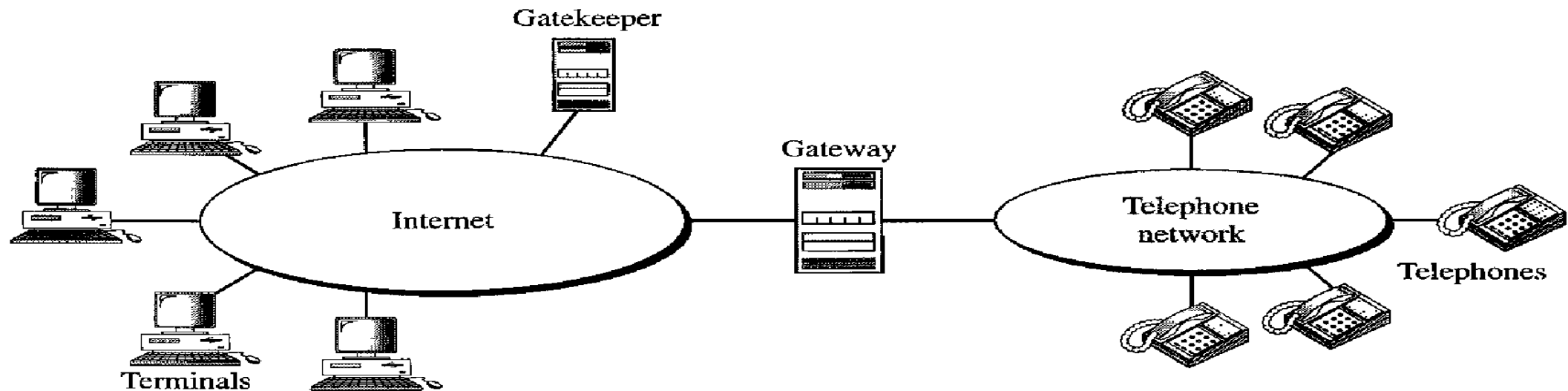


# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

## H.323

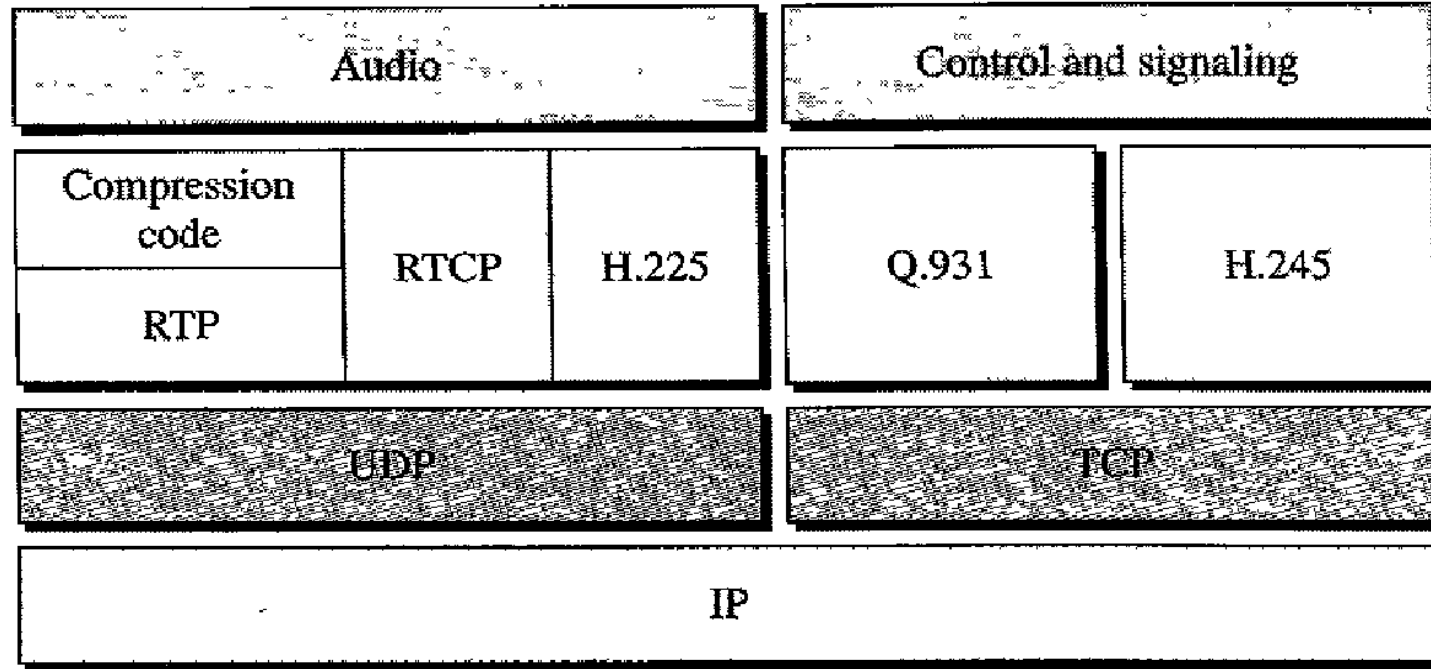
**H.323** is a standard designed by ITU to allow telephones on the public telephone network to talk to computers (called *terminals* in H.323) connected to the Internet. Figure 29.25 shows the general architecture of H.323.

**Figure 29.25** *H.323 architecture*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 29.26 *H.323 protocols*



# Connexion à distance, courrier électronique et transfert de fichiers, WWW et HTTP

Figure 29.27 H.323 example

